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THE
SYNTHI
EDUCATIONAL
HANDBOOK

by
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February 1972

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EMS
Electronic Music Studios
SYNTHESIZERS - VOCODERS

USING THE EMS SYNTHI A, AKS AND VCS3 WITH A MIDI TO CV CONVERTER

Although the EMS synthesizers were designed well before the advent of MIDI and lack actual CV and Gate input and output sockets it is quite simple to adapt them for use with a MIDI to CV system.

CONNECTING THE CV INPUTS

All Synthis have two HI-LEVEL INPUT CHANNELS, which can function as line level signal inputs AND Control Voltage (CV) Inputs. This means that two channels of CV information can be connected simultaneously and patched to control any function on the matrix - not just pitch.

The only complication in connecting CVs to these inputs is that due to the high gain of the input amplifiers it is necessary to include a 100K resistor in line (in series) with the incoming CV. This is best done by making up two special quarter-inch jack leads with resistors incorporated and labelling them to avoid mixing them with normal jack-leads.

CONNECTING THE GATE INPUT

ALREADY MODIFIED

The Gate input on an unmodified Synthi is only accessible on Pin 5 of the 8-way KEYBOARD socket. A ~~simple modification~~ is recommended which should internally link across from this pin to the adjacent CONTROL OUTPUT CH2 socket (via a diode), so that the Gate Input is available on a quarter-inch jack.

Connections to this Gate Input should be made with a normal jack lead, and not the special 100K resistor lead described above.

A moderately competent technician should proceed as follows to make this modification:

- 1) Remove Synthi from case or remove back and bottom panels if a VCS3.
- 2) Remove two retaining screws from the Reverb spring box - if a suitcase synthi, and rest spring unit on bench. The thin blue and red wires are very delicate so try not to wrench them off.
- 3) Locate Pin 5 at the back of the keyboard socket. This is normally a yellow wire on later machines, but the pins are numbered on the black plastic housing - use a bright lamp if need be.
- 4) Pull black sleeve back up the wire to expose soldered terminal. Use

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a soldering iron to detach this yellow wire temporarily.

- 5) We want to have two wires coming from this pin 5 so join another short length of wire - say 20 cms of yellow 7/0.2, reconnect to pin 5 and replace sleeve.
- 6) Snip off the thin wire going to the Control Output Ch2 jack 'hot' terminal, and insulate with tape or a sleeve. This is now abandoned.
- 7) Connect a small signal diode - a 1N4148 for example - to this 'hot' terminal. The dark polarising band must be at the opposite end from the jack terminal.
- 8) Connect the end of the diode with the polarising band to the free yellow wire end and sleeve over the diode.
- 9) Re-label the Control Output Ch2 socket as 'EXT GATE INPUT'.

Alternatively a plug-in harness can be constructed with a male 8-way keyboard plug and jack socket, but this involves more parts and labour as well as being more untidy in use.

If you have any problems or need any further advice please contact Robin Wood on 0726-883265 (UK).

THE SYNTHI EDUCATIONAL HANDBOOK

Wee also have Sound-houses, where wee practise and demonstrate all Sounds, and their Generation. Wee have harmonies which you have not, of Quarter Sounds, and lesser Slides of Sounds. Diverse Instruments of Musick likewise to you unknowne, some sweeter than any you have; Together with Bells and Rings that are dainty and sweet. Wee represent Small Sounds as well as Great and Deepe; Likewise Great Sounds, Extemuate and Sharpe; Wee make diverse Tremblings and Warblings of Sounds, which in their Originalle are Entire. Wee represent and imitate all Articulate Sownds and Letters and the Voices and Notes of Beasts and Birds. Wee have certain Helps, which sett to the Eare doe further the Hearing greatly. Wee also have Strange and Artificiall Echos's, Reflecting the Voice many times, and as it were Tossing it: And some that give back the Voice lowder than it cam, some Shriller, some Deeper; Yea some rendering the Voice, Differing in the Letters or Articulate Sound, from that they receyve, Wee have also means to convey Sounds in Trunks and Pipes, in strange Lines, and Distances...

Roger Bacon "The New Atlantis" 1624

USER NOTES FOR EMS DYNAMIC KEYBOARDS - CRICKLEWOOD/DK1/DK2

These keyboards are all 3-octave, C-to-C, with an internal sawtooth oscillator and velocity sensing contacts. The DK2 varies in that it produces a duo-phonic output, sensing the lowest note and highest note played, while the DK1 and Cricklewood are monophonic high note priority only.

The only difference between the earlier Cricklewood and the DK1 which replaced it - apart from its charming name - is the finish of the cabinet. The Cricklewood is entirely of french-polished afromosia hardwood, whereas the DK1 (and DK2) has afromosia end-cheeks with black vinyl-covered plywood elsewhere.

These keyboards can all be used with any EMS VCS3 or Synthi A synthesizer, but cannot be used on their own. An 8-way umbilical cable connects the keyboard to the synthesizer.

CONTROLS

Six keyboard controls are provided:

Frequency	This shifts the tuning of the keyboard's internal sawtooth oscillator to give a total range of approximately 30 to 2000 Hz.
Tuning Spread	This trims the tuning of the internal oscillator over a narrow range, allowing the octave to be 'stretched' or 'compressed'. Use the middle F sharp as the centre reference for tuning, tuning up or down from this point.
Level	This control sets the signal output level of the internal oscillator. (Note: When an external signal input is connected to the jack socket on the rear panel, this control functions so as to mix in the keyboard's internal oscillator signal with the incoming signal.)
Dynamic Range	The Dynamic Range control determines the degree of variation in loudness of the internal oscillator (and keyboard input) in response to key velocity. At 0 the loudness will remain constant and independent of key velocity. At maximum the loudness will vary by more than 40 db depending on the player's touch.
(Input Ch. switches)	As there are only two input channels on the VCS3/A and the keyboard produces four outputs, these switches are needed to select which outputs are used.
Input Ch1 switch	Input Channel 1 switch can select between Signal - the keyboard's internal sawtooth oscillator - or Keyboard Voltage. In the case of the DK2 this is the high-note voltage. The Signal or Keyboard Voltage is then routed to row 8 on the Synthi A/VCS3 patchboard.
Input Ch2 switch	Input Channel 2 switch can select between the key velocity voltage - Dynamic Voltage - and the second Keyboard Voltage. In the case of the DK2 this is the low-note voltage.

THE JACK SOCKET

This socket is found on the back panel next to the 8-way connector. Its function is to provide access for an input signal to the keyboard oscillator's dynamic modulator. The loudness of this input signal can then be varied by key velocity.

In the Whole Earth Catalogue 1970 Walter (now Wendy) Carlos gave a damning review of the VCS3, condemning it as a 'real toy' on the principal grounds that when the Dynamic Voltage is patched to control the output VCA - pin O9/P9 - the result is that a light keyboard touch results in a loud signal and vice-versa. This is true, but had Carlos investigated further he might have discovered that any signal patched to the Meter column on the VCS3/A matrix is automatically routed to the keyboard's signal input. It may then be mixed with the keyboard oscillator signal, but more importantly it can be dynamically modulated by key velocity according to the Dynamic Range control setting. The whole output then re-appears on Row 8. Perhaps Carlos' judgement was piqued by the fact that even on his monster modular Moog there was no keyboard velocity output.

Returning to serious matters then, remember that the keyboard modulator input can be accessed directly from the Meter row on the matrix as well as by the jack socket on the keyboard.

OPERATION

The envelope shaper on the VCS3/A is triggered from keys pressed on the DK keyboard, and this connection is permanently made by the 8-way umbilical cable.

To use the keyboard's internal oscillator, connected to Output Ch2 via the VCS3/A envelope shaper, place pins at: D8 and C12, with the keyboard's Input Ch1 switch set to Signal. The oscillator loudness is controlled by both the Level control on the keyboard as well as the Input Level control for Channel 1 on the VCS3/A. Use the Frequency control to set the required pitch range and trim the Tuning Spread control for correct intonation by tuning from the middle Fsharp either up or down. Set Dynamic Range to 0 at first, until the basics have been mastered.

To control the VCS3/A's own oscillators it is necessary to select one of the Keyboard Voltages on the 'Input Ch' switches. If Input Ch2 Keyboard Voltage is used this allows the Keyboard Oscillator to be used on Input Ch1 and create a thicker sound. A simple patch using the envelope shaper and Oscillators 1 and 2 would consist of pins: I9, J9, D3, D4 and C12. To add the Keyboard Oscillator simply add a pin at D4. Set the keyboard Input Ch1 switch to Signal, and Input Ch2 switch to Keyboard Voltage.

Tuning the VCS3/A oscillators in this patch is not at first a very straightforward matter, but as with most things practice will help considerably. The octave spread is set on the Input Level Channel 2 control, which should be set at approximately 6 to begin with. Turn the Level of the Keyboard Oscillator to 0 and the output levels of one of the two VCS3/A oscillators to 0, to leave just one of the VCS3/A oscillators sounding when a key is pressed. Next identify the note on the keyboard which moves least in pitch when the Input Level Channel 2 control is moved. This should be middle F sharp for a DK1 or Cricklewood, and possibly near the C above with a DK2. Now tune from this note, varying Input Level Channel 2 control until the correct tuning is reached. Increase the level controls of the other oscillators, tune their frequencies together and see what the keyboard sounds like across its

range. Repeat this process until tuning is optimised. Note: Red pins should be used for pins I9 and J9 if available.

To make the loudness of all three oscillators respond to the key velocity the pins D3 and D4 should be moved to B3 and B4 and the Dynamic Range control set at about 6. The composite output of the three oscillators will now appear at Row 8. Note: With a DK2 whenever more than one key is held down at a time the response becomes duo-phonic as in this sample patch. Note: With a Mk1 VCS3/A the two pins would need to move to A3 and A4.

To experiment with the Dynamic Voltage make the following patch: D8, H12, C10 and N9. Select Dynamic Voltage and Signal, and set the Filter Response control to sharply resonant. Set both Input Level Channel controls to 10. Note how a varying keyboard touch radically alters the Filter Frequency and thereby the overall timbre. Note: Playing with an overlapping (legato) style will preserve the existing Dynamic Voltage level.

Note: Sometimes the Keyboard Oscillator can be heard wailing away in the background when it is not wanted. A blank mono jack-plug inserted in the Keyboard Modulator input jack will normally cure this.

Note: The umbilical keyboard connecting lead is not simply wired pin-to-pin. It contains two 100K resistors in series with Pin 1 (Input Ch1) and Pin 6 (Input Ch2). These are gain-setting resistors and must not be omitted.

If you have any problems with your DK keyboard please contact Robin Wood on 0726-883265 for assistance.

External Connections:

6	5



OUTPUT FILTER

CHANNEL 1 - - - OUTPUT - - CHANNEL 2



External Patch:

out chan	1
----------	---

2.	
----	--

05C 1

10	0.00
750	0.00

2	1
0.50	0.50

3	7
noise	

1	noise
---	-------

inputs	1
--------	---

filter	7
--------	---

plaza	plaza
plaza	plaza

lead is not
trapezoid

ring mod	
ring sig	

1000000

reverb

stick	↑
	→

[illegible]

A B C D E F G H I J K L M N O P

SYNTHI Dopesheet E.M.S.(London) Ltd.		Description:		External Connections:			
Name : Sheet : Patch : Start : End : Notes :	<table border="1" style="border-collapse: collapse; text-align: center;"> <tr> <td style="width: 20px; height: 20px;">C</td> <td style="width: 20px; height: 20px;">S</td> </tr> </table>		C	S	FILTER/OSC RINGMOD 		
C	S						
OSCILLATOR 1		ENVELOPE SHAPER					
OSCILLATOR 2		OSCILLATOR 3		REVERBERATION			
NOISE GENERATOR		OUTPUT FILTER		INPUT LEVEL			
CHANNEL 1 -- OUTPUT -- CHANNEL 2				RANGE			

External Patch:	SIGNALS						CONTROLS									
	1 out	2 out	env	A r.m.	B	reverb	1 filter	1 osc	2 freq	3	decay	reverb	1 filter	2 out lev		
out chan	1	2													1	2
osc 1															3	4
osc 2															5	6
osc 3															7	8
noise															9	10
inputs															11	12
filter															13	14
trapezoid															15	16
env signal																
ring mod																
reverb																
stick																
↑																

A B C D E F G H I J K L M N O P

SYNTHI EDUCATIONAL HANDBOOK

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INTRODUCTION

Primarily, this book is intended to show you how to use your SYNTHI AKS. Although it is not a comprehensive text-book, it may also tell you something about the way in which sounds are made or altered.

The SYNTHI AKS is a *Synthesiser*, which means that it is used to build up or *synthesise*, sounds from basic tones. It differs from other musical instruments in that it has no characteristic sound of its own, and in order to use it you have to know something about the nature of sounds so you can build your own creatively. For this reason, we do not apologise for writing a book which is educational in tone. We hope that this book will be used in schools which have a SYNTHI AKS, and we also hope it will be used by others, in every field, to familiarise themselves with what a sound is before they become confused by a multitude of electronic devices.

In the first chapter, USING YOUR SYNTHI, we explain how the SYNTHI is set up and what the controls do. Unless you have experience of electronic equipment, you will probably not understand much of this chapter, but you will find later on when you have learnt a bit more about the SYNTHI, that it is a useful chapter for reference.

In the other chapters, there are sections of explanation and experiments. (The experiments are written in italics for clarity). The experiments are not intended to be complete: rather they are starting points for the much more detailed experiments which we expect you to do. When you buy a constructional toy, you may at first make the models shown in the instruction book, but you will soon learn the general principles and make models of your own which express your own interests. We want you to think of the SYNTHI as a construction set, with which you can build your own sounds. The "patches" we provide are not complex, because we want to illustrate how the SYNTHI works, but when you have understood an idea or technique, you should try and incorporate it into your own patches.

There is not much mention of oscilloscopes in this book because most SYNTHI owners will probably not have one. But many (perhaps most) of the experiments are enhanced if an oscilloscope is connected to the SYNTHI and you should do this if possible.

CHAPTER 1: USING YOUR SYNTHI

This chapter describes the use of the SYNTHI, and also describes the devices it contains. It should be read through first of all (it may seem rather abstract initially) and then should be used for reference later, when you are doing the experiments for the SYNTHI. Before doing any of the experiments, make sure you have read the next section, Connecting the SYNTHI.

Unlike most of the electronic devices we are accustomed to, which perform one function only (radio and television receivers for example), or a few functions (a radiogram can play records and receive radio programmes), the SYNTHI has very many functions. The SYNTHI contains a number of separate circuits which are connected by the user; the knobs on the front control the circuits and the patchboard (the black square in the lower centre) connects them. If the SYNTHI looks complicated to you, remember that it uses only simple devices, and that complex sounds arise from the use of several devices together.

Connecting the SYNTHI

Power Supply

Before it can be used at all, the SYNTHI must be connected to the mains. European models are adjusted for 200-250VAC, and North American models for 100-120VAC. If for any reason the setting is incorrect it should be altered, using the slide-switch on the front panel before connecting the SYNTHI. The SYNTHI uses a transformer designed for 50-60Hz AC *and must never be connected to a DC supply.*

Connect the SYNTHI to the mains supply with the lead provided. The earthing connection (green/yellow) is not essential and may be left unconnected if two pin sockets only are available. Turn the POWER switch on and check that the mains indicator light glows red.

Signal and Control Connection

Other electronic equipment is connected to the SYNTHI by the jacks at the top of the panel. The SYNTHI can be used as an entirely self-contained instrument, but its resources can be exploited more fully when other equipment is used with it. In particular, the built-in loudspeakers are very small and, although they are adequate for monitoring purposes, an external amplifier and loudspeaker system should be used for anything but personal experiments. The functions and specifications of the jacks are explained here.

<u>Stereo Phones:</u>	Stereo (three wire) jack giving a two channel signal suitable for powering a pair of stereo headphones.
<i>Level:</i>	<i>10V p-p max into 50 ohms</i>
<u>Scope:</u>	A mono Output (two wire) connected to the Meter Circuit. It is suitable for an oscilloscope display, and is sometimes connected to the SYNTHI inputs as part of a patch.
<i>Level:</i>	<i>Device dependent, 5V p-p max</i>
<u>Signal Outputs:</u>	These outputs are suitable for a power-amplifier or stereo tape-recorder.
<i>Level:</i>	<i>2V p-p max into 600 ohms</i>
<u>Control Outputs:</u>	These outputs should be used when the SYNTHI is required to provide a DC control voltage to another synthesizer or voltage controlled circuit. They are connected to the Output Channel rows on the Patchboard.
<i>Level:</i>	<i>+5VDC max into 10Kohms</i>
<u>Keyboard:</u>	This socket is designed to connect the SYNTHI to a SYNTHI 'K' or 'KS' keyboard and should not be used for any other purpose.

Signal Input: Jacks connected to the Input Amplifiers of the SYNTH1. An external source with line-output, such as a pre-amplifier or tape-recorder, should be connected here.

Sensitivity: 2.5VAC p-p
+3.5VDC into 50Kohms for voltage-control

Microphone Input: These inputs are suitable for most forms of low-level signal, as produced by air-microphones, contact microphones and guitars.

Sensitivity: 5mVAC into 600ohms

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The Devices

The SYNTHI has a number of devices, each with its own controls on the front panel. The brief specifications below are intended for reference, and do not describe the uses of the device.

Control refers to the function of the device, if any, which can be voltage-controlled. *Matrix* gives the identification by digits and letters of the rows and columns on the Patchboard allocated to the device.

Oscillator 1: Audio frequency oscillator with sine and ramp waveforms. Manual controls for frequency, shape of sine output (odd harmonics may be added), level of sine output, level of ramp output. The outputs are mixed at the Patchboard.

Range: 1Hz - 10KHz on dial
Output: sine 3V p-p
ramp 4V p-p
Control: frequency 0.132 V/octave
Matrix: 3, I

Oscillator 2: Audio frequency oscillator with rectangular waveforms. Manual control for frequency, shape, levels. Outputs are mixed at the patchboard.

Range: 1Hz - 10KHz on dial
Output: square 4V p-p
triangle 3V p-p (symmetrical position)
ramp 6V p-p (extreme position)
Control: frequency 0.32 V/octave
Matrix: 4, J

Oscillator 3: Audio and subsonic oscillator with rectangular and triangular waveforms. Manual controls for frequency, shape and levels. This oscillator may be run at very low frequencies for control applications.

Range: 0.02Hz - 500Hz on dial
square 4V p-p
triangle 3V p-p (symmetrical position)
ramp 6V p-p (extreme position)
Control: frequency 0.26 V/octave
Matrix: 5, 6, K

Noise Generator: White and coloured noise source. Manual control of level and colour.

Output: 3V p-p

Filter/Oscillator: A multi-purpose filter, resonator and oscillator, with adjustable bandwidth (Q) and frequency. Manual control of frequency, mode and output level.

Range: 5Hz - 10KHz on dial
Cut-off Rate: 18db/octave max
Q: 20 max
Control: frequency 0.2 V/octave
Matrix: 10, H, N

Ring-Modulator: An advanced design with high input rejection. Manual control of output level.

Input Level: 1.5V p-p max
Output: 6V p-p max
Rejection: -60db at 1.5V input
Matrix: 13, E, F

>

Envelope Shaper: A programmed variable-gain amplifier for envelope control. Manual controls for attack, on, decay and off times, signal output level and control output ("trapezoid") level. There is also a recycle ("attack") button by the Joystick.

Timing: Attack 2ms - 1s
On 0 - 2.5s
Decay 3ms - 15s
Off 10ms - 5s + "off" position to inhibit recycling.

Signal Output: 5V p-p max

Control Output: +3VDC max

Control: Decay time 0.4 V/octave

Matrix: 11, 12, D, L

Reverberation Unit: A double spring device with delays of 25ms and 30ms. Due to the small cabinet of the SYNTH1, acoustic feedback can occur when the internal loudspeakers are used and this should be avoided as far as possible. Manual controls for mix and level.

Max Reverb Time: 2s

Output: 5V p-p max

Control: Reverberation time -2V for 0%, +2V for 100% mix

Matrix: 14, G, M

Joystick: Two manual control voltages are provided by moving the joystick up or down or from side to side. Manual range controls are provided.

Output: +2VDC

Matrix: 15, 16

Meter: Dual purpose meter for monitoring, checking and cueing. Manual control for selecting signal or control modes.

Input: +1VDC in Control Voltage mode

4V p-p in Signal Level mode

Input Amplifiers: Signals and controls may be input to the SYNTHI at line or microphone level. Manual control of sensitivity.

Sensitivity: 2.5VAC signal
+2.5VDC control
5mVAC microphone

Matrix: 8, 9

Output Amplifiers: Two output channels with manual control of level, filtering and panning. Internal monitoring speakers with muting switch.

Outputs: 2V p-p into 600 ohms signal panned
10V p-p into 50 ohms signal unpanned
+5VDC into 10Kohms unpanned

Control: level 15db/V

Matrix: 1, 2, A, C, O, P

Signal Trigger: A signal sent to Channel 2 can be used to trigger the Envelope Shaper if the Channel 2 muting switch is set to the mid-position, "Trigger".

Keyboard: 32 Touch Contacts providing pitch and Dynamic Output. Sequences with 256 event storage, and touch controls for Record, Play, instant Transposition and random note generation.

Matrix: 8, 9, 16

Patching

The devices in the SYNTHI are not connected internally. Before the SYNTHI will make any sound at all it must be "programmed" by putting pins into the Patchboard. Each pin connects the output of one device to the input of another. Suppose you want to connect the output of the Noise Generator to the input of the Filter, for example. Look down the labels at the left hand side of the Patchboard and you will see that row 7 is labelled "noise". Now look along the top of the Patchboard to "filter" at the top of the 8th column: putting a pin at the intersection of the 7th row and 8th column will connect the two devices as required. It helps to imagine the signal travelling from left to right along the 7th row until it meets the pin, and then travelling up to the top of the Patchboard along the 8th column.

Later in this book, patches will be described by a letter A - P and a number 1 - 16. These refer to the column and row respectively and for convenience they are inscribed on the right and bottom sides of the Patchboard. The pin used in the example above (Noise to Filter) was H8.

The 32-way socket in the spare pin repository at the bottom of the panel is connected to each row and each column of the Patchboard. It is used for "Prestopatches" which set up any combination of pins immediately.

CHAPTER II: SIMPLE PROPERTIES OF SOUNDS

We are very accustomed to hearing sounds. Animals, people, machinery and natural events make sounds. People in particular make complex sounds to communicate with each other, and go to great lengths to make instruments which produce the organised sounds called music. Anything that makes a sound we will call a *source* of sound.

What do sources of sound have in common? The answer is simply that they *move*. There are many different kinds of movement and they give rise to many different kinds of sound. Some movements are obvious - a rock falling, or waves on the beach. But many sounds are caused by tiny vibrations which are not visible to the unaided eye, such as the vibrations of a tiny bell when it is ringing.

The reason that we hear sounds is that the movement of the sound source makes the surrounding air move, and it is the air movement that our ears detect. Inside each ear there is a small mechanism which converts vibrations of the air into vibrations of a nervous membrane and then into electrical vibrations which are interpreted by the brain.

During the last hundred years, men have developed their own artificial ways of turning sound vibrations into electrical vibrations and *vice versa*. It was by doing this that they were able to send sounds over long distances by using telephones and radios, and to store them in permanent form on gramophone records and recording tapes. A *microphone* is a device which turns sounds into electrical signals, and a *loudspeaker* turns electrical signals back into sounds.

Loudness

Loudness is perhaps the most obvious quality of a sound. The amount of extra pressure caused by the vibrations on your ear-drum determines loudness: a sound that you can only just hear has a sound pressure of 0.00002 N/m^2 (Newtons per square metre - don't worry if you have not heard of this unit of pressure before, we are only using it for comparisons) and the sound of a rifle shot as heard by the marksman has

a sound pressure of about 2000 N/m^2 . The astonishing thing about these two figures is that one is a hundred million times larger than the other: It is as if we were to make a weighing machine that could weigh fleas as accurately as elephants!

There are two reasons why we do not normally use a unit of pressure to measure sound. One is the obvious inconvenience of using numbers which can vary over such a wide range, and the other is that it does not agree with our own idea of loudness. An orchestra playing loudly produces a thousand times as much sound as the same orchestra playing softly, and a million times as much sound as a quiet conversation: but it seems that the ear responds not directly to the power, but to its logarithm. In other words it is *ratios* of sound power that we respond to.

Using a logarithmic scale, we find the numbers are much more reasonable. In the table, Fig. II-1, the unit is a *decibel* (db) and since the decibel is a ratio we say that sounds are measured in terms of the quietest sound that can be heard, which is consequently 0 db on the scale.

<i>Quietest sound</i>	<i>0 db</i>
<i>Gentle rustle of leaves</i>	<i>10 db</i>
<i>Whisper at 4 feet</i>	<i>20 db</i>
<i>Quiet bedroom</i>	<i>35 db</i>
<i>Conversation at 12 feet</i>	<i>50 db</i>
<i>Busy office with typewriters</i>	<i>65 db</i>
<i>Alarm clock at 3 feet</i>	<i>80 db</i>
<i>Heavy diesel lorry at 20 feet</i>	<i>90 db</i>
<i>Very noisy factory</i>	<i>100 db</i>
<i>Jet aircraft taking-off at 75 feet</i>	<i>140 db</i>
<i>Moon rocket taking-off at 1000 feet</i>	<i>200 db</i>

Fig. II-1: Table of sound levels

Experiment 1

Turn the SYNTHI on and connect it (if possible) to power amplifiers. Connect Oscillator 1 to Output Channel 1 by putting a pin at B1. (If you don't understand this, re-read the section on "Patching" in Chapter 1.) Set the FREQUENCY control of Oscillator 1 to 7, the SHAPE control to 5, and the LEVEL controls to 8 (sine) and 0 (ramp). Set OUTPUT LEVEL 1 to 7 (or to a convenient listening level) and check that OUTPUT FILTER 1 is set to 5 (centrally).

By altering the LEVEL control, observe that the sound can be made inaudible or (if sufficient amplification is available) painfully loud. By altering the FREQUENCY control note that the level at which the sound becomes inaudible depends on the frequency.

As well as being loud or soft, sounds can be long or short. Some sounds are continuous - their loudness is the same all the time. Other sounds start and stop; they are called *transient* sounds, and the exact way in which they start and stop is very important in music, which is made up almost entirely of transient sounds.

Experiment 2

Connect Oscillator 1 through the Envelope Shaper to Output Channel 1. The input to the Envelope Shaper is column D and its output ("env signal") is row 12, so the pins required are at D1 and A12. Make sure that the Oscillator and Output Controls are set to give an audible signal, and turn the Envelope Shaper SIGNAL OUTPUT control to 8.

Set the timing controls of the Envelope Shaper (labelled ATTACK, ON, DECAY, OFF) according to the table, Fig. II-2, and listen to the different envelopes. (a) has no attack or decay, (b) has a decay but no attack, (c) has both an attack and decay. (d) is the same as (c) but it is necessary to push the "attack" button by the Joystick to start a new cycle. Experiment with the timing controls yourself, trying perhaps to imitate various instruments. Alter the frequency of the Oscillator and listen to the different aural effect of the same envelope at different pitches.

Add a pin at B11 and set the Meter switch to CONTROL VOLTAGE. The output at row 11 is not a signal but a control voltage (of which more later) - and in this case it is the voltage used to control the signal output level of the Envelope Shaper. The Meter should move in time with the Attack/Decay cycle - if it doesn't, turn up the TRAPEZOID LEVEL control. Fig. II-3 shows the way in which the loudness of the sound coming from the Envelope Shaper changes with time, and it also represents the voltage coming from the Trapezoid, which explains its name.

Setting:	(a)	(b)	(c)	(d)
ATTACK:	0	0	4	4
ON:	3	3	2	2
DECAY:	0	7	7	7
OFF:	5	0	0	10

Fig. II-2: Envelope Shaper timing control settings for
Experiment 2

Frequency

The second obvious quality of a sound is its *frequency*. This is both more complex and in many ways more important than loudness. More complex because most sounds have more than one frequency, and more important because we identify sounds more by their frequency structure than by their loudness. Frequency is the *rate of vibration* of the sound source, measured in cycles per second, which we abbreviate to Hertz (Hz).

We hear different frequencies rather as we hear different loudnesses, with a logarithmic law. When the frequency of a sound is doubled, we hear it an octave higher. 800Hz is an octave higher than 400Hz and two octaves higher than 200Hz. We can hear sounds over almost ten octaves, from 20Hz to 20000Hz, and the ear is most sensitive at 4000Hz. As we get older, our ability to hear very high sounds diminishes and only young people can hear sounds above 20000Hz.

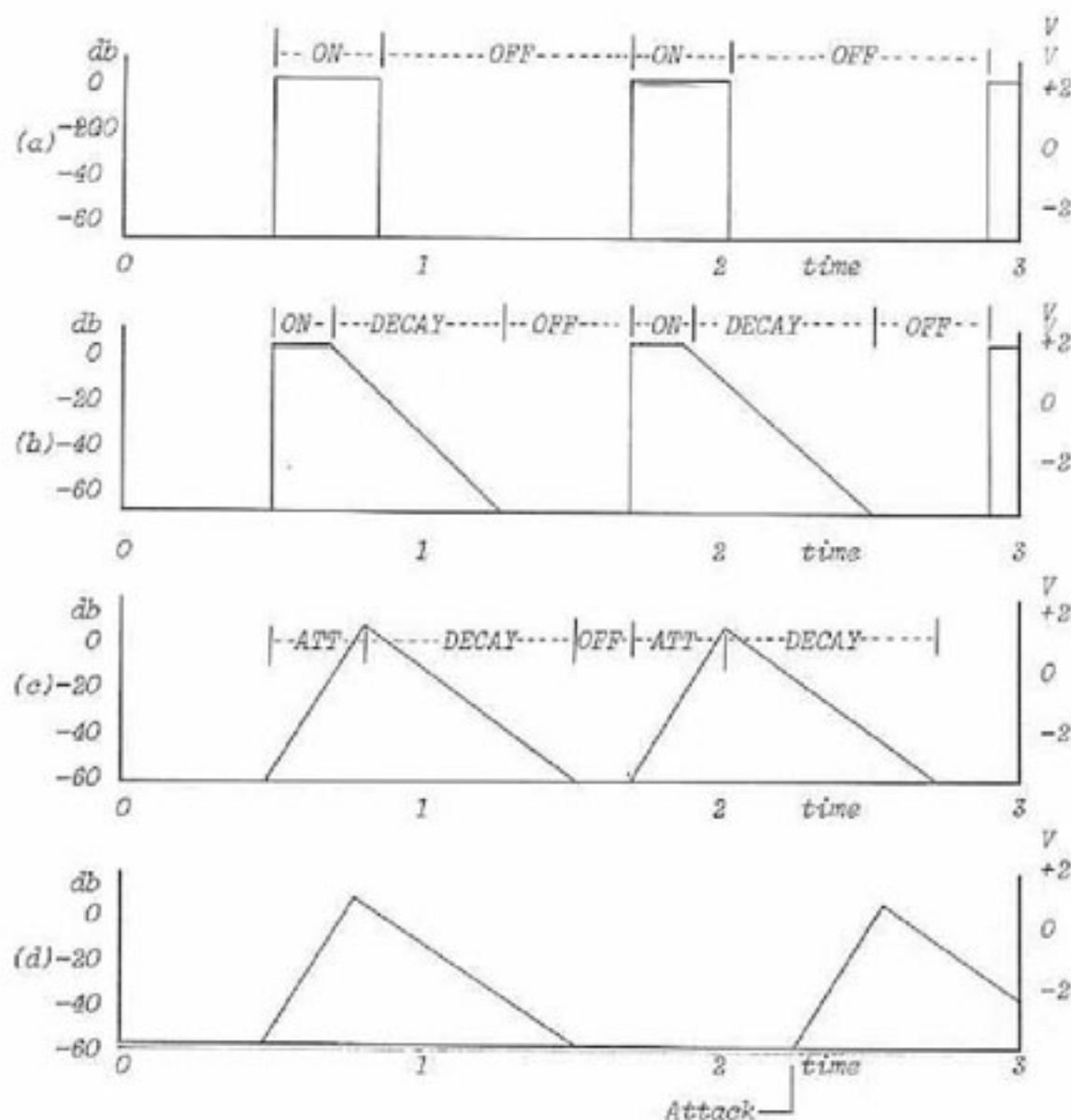


Fig. II-3: The Envelope Shaper

The graphs show time horizontally and output signal vertically. The left hand vertical scale is db measured with 0 db as maximum loudness, and the right hand scale is the trapezoid voltage. See Fig. II-2 for the control settings to produce (a), (b), (c) and (d).

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Experiment 3

Use the same experimental set-up as in Experiment 1. Set the SHAPE control to 5.

Adjust the RAMP LEVEL control to zero, and the SINE LEVEL control for a fairly loud signal, and move the FREQUENCY control. At very low frequencies the sound will disappear altogether (or it may turn into a series of clicks which are heard as separate sounds when they are slower than about 20 per second) and at very high frequencies it becomes very thin and then disappears. (It may be impossible to make the frequency high enough using the dial alone. If this is so use the Joystick to extend the frequency range as follows: put a pin at I16 and set the VERTICAL RANGE control to 10, then move the Joystick up and down to control the frequency.)

Most sounds contain not one frequency but several, and we talk about frequency components.

Experiment 4

Use the same experiment set-up as Experiment 2.

With the SHAPE control at 5, the sound from Oscillator 1 is almost "pure". Turn the SHAPE control either way and note that the sound gets "buzzier" in quality. By moving the FREQUENCY control try to discover whether your range of hearing is altered by the change in tone. Move the pin to B2 and listen to the different qualities of "square" and "ramp" waveforms, adjusting the SHAPE control and both LEVEL controls of Oscillator 2.

The "buzzier" quality of the sound in this experiment is due to the presence of extra frequencies. The "pure" sound is a vibration at a certain frequency, f say, and the SHAPE control introduces other frequencies at $2f$, $3f$, $4f$, etc. Provided that the new frequencies are simple multiples (twice, three times, four times, etc.) of the original

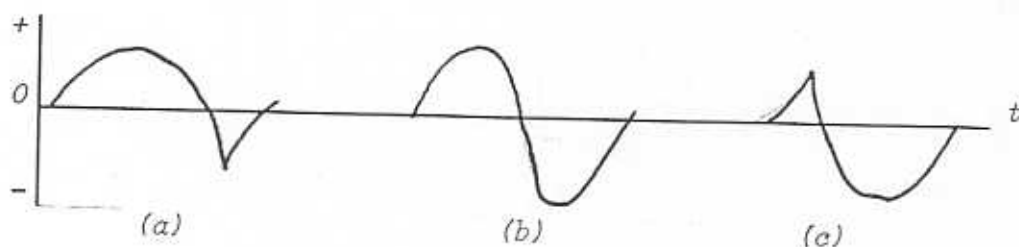


Fig. II-4: Effect of Sine SHAPE Control

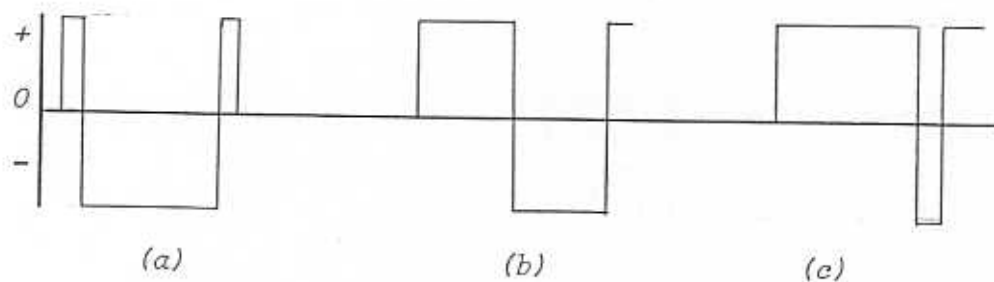


Fig. II-5: Effect of Square SHAPE Control

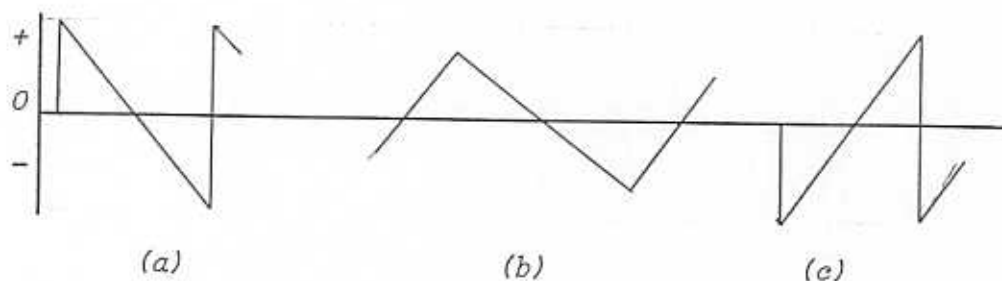


Fig. II-6: Effect of Triang SHAPE Control

In Figs. II-4 thru II-6, (a) corresponds to a SHAPE settings of 1, (b) to a SHAPE setting of 5, and (c) to a SHAPE setting of 9.

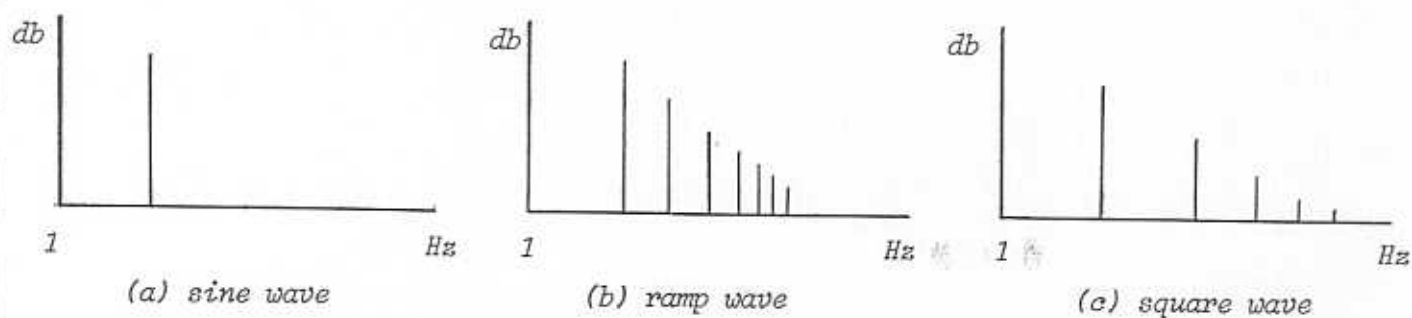


Fig. II-7: Spectra of three waveforms

frequency, we hear only a single note with a characteristic tone. The initial frequency (f) is called the *fundamental* and the other frequencies are called *harmonics* or *overtones*. The "tone" of the note is technically called its *timbre*. In the experiment above, you should have found that you could hear the Oscillator at lower frequency settings when the SHAPE control was moved, since although the fundamental became inaudible (less than 20Hz) the overtones could still be heard.

We will now describe two ways in which a sound can be represented on a graph. Firstly, we can draw the sound as it appears on an oscilloscope screen. What we are doing here is plotting a variable which changes with time: it does not matter much whether we plot air-pressure, movement of the ear-drum, position of the diaphragm of a microphone or the voltage which the microphone produces, because they all look the same. The voltage is most convenient, because the oscilloscope is an instrument designed especially to plot a changing voltage. Figs. II-4 to II-6 show the waveforms that we have heard so far as they would appear on an oscilloscope screen. These drawings clarify the meaning of the little squiggles that you can see on the front panel of the SYNTHI by the Oscillator controls, and also explain the names "square" and "ramp" which we have given to the waveforms. By listening to the sounds and looking at the graphs you will learn to associate a sound with its waveform: note that the waveforms with the sharpest corners have the hardest sounds. If you have access to an oscilloscope, connect it to the SYNTHI output or SCOPE jack and experiment with the SHAPE controls while watching and listening.

The other way in which a sound is represented on a graph is slightly more difficult to understand, but it is often more useful. Horizontally we use a *frequency* scale and vertically we use an *amplitude* scale. Since there is no time axis, this sort of graph represents a continuous sound. For reasons which you will now understand, both scales in this graph are logarithmic: in fact, it is convenient to use octaves for the frequency scale and decibels for the amplitude scale. The resulting graph is called the *spectrum* of the sound, and you will see that the spectrum of a sound is analogous to the spectrum of light, since we see different frequencies of light as different colours.

Now we will look at the spectra of some of the sounds we have been listening to; see Fig. II-7. (a) shows the spectrum of a pure sound which has exactly one frequency component, represented in the graph as a single line at that frequency (we have chosen 160Hz). (b) shows the spectrum of a ramp waveform at the same fundamental frequency (160Hz) but now there are overtones at the frequencies of the harmonics (320Hz, 480Hz, etc). The harmonics are equally spaced in frequency (each being 160Hz higher than the last) but appear unequal on the logarithmic scale of the spectrum. They also sound unequal musically: the intervals between harmonics are an octave, a fifth, a fourth, a third and so on, and this justifies our use of a logarithmic scale. (c) shows the waveform of a square wave which has odd harmonics only ($3 \times 160 = 480\text{Hz}$, $5 \times 160 = 800\text{Hz}$, $7 \times 160 = 1120\text{Hz}$, etc). This is a property of all symmetrical waveforms: if they are the same as their mirror-image, there will be no even harmonics.

There is an instrument which displays the spectrum of a sound directly, rather as an oscilloscope displays the waveform, but it is complicated and expensive. However, the SYNTHI can be used to demonstrate the frequency structure of a sound acoustically. A *filter* is a device which allows some things to pass through it but blocks others. An oil filter is a wire mesh which allows oil to pass through it but stops dirt, and a food strainer allows water and dirt to pass through it but stops vegetables. In sound experiments, a filter is used to select or reject certain frequency components of the sound. Acoustic filters can be made, but nowadays it is more convenient to filter an electrical signal and listen to the result with a loudspeaker.

Experiment 5

Connect the Filter to Output Channel 2 with a pin at C10. The Filter should be set to give the highest possible selectivity without resonating as follows: turn the RESPONSE control to about 5 and the LEVEL control to 10. Turn the FREQUENCY control all the way from 1 to 10; if there is no sound at all, turn the RESPONSE up a bit and sweep the FREQUENCY again. Do this until the Filter resonates (makes a pure sound of its own: there is no input signal yet) over a small

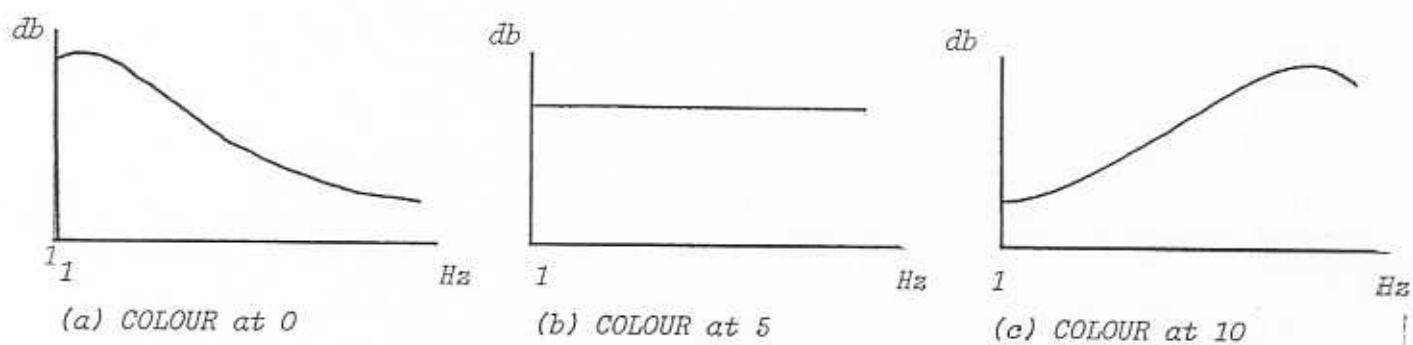


Fig. II-8: Effect of COLOUR Control on Noise Generator

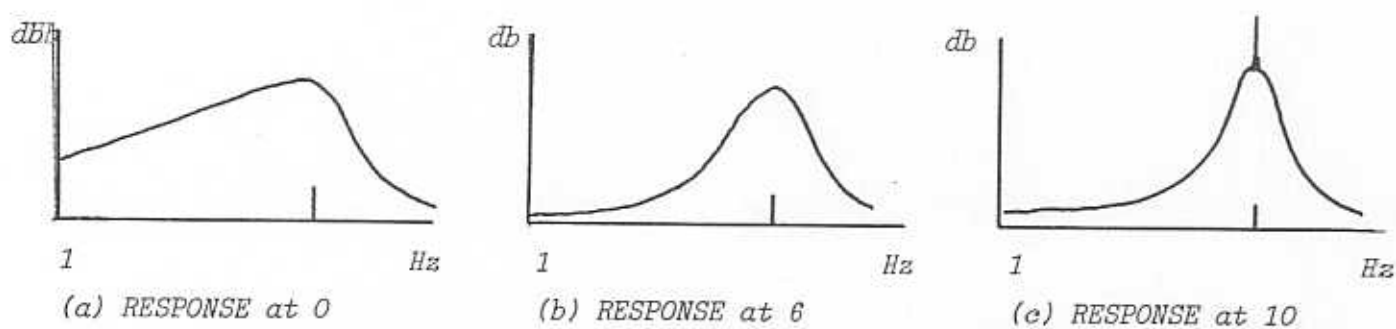


Fig. II-9: Effect of RESPONSE Control on Filter/Oscillator

frequency range, and then turn the *RESPONSE* down a bit. The Filter is now in bandpass mode, which is to say that it will only let through a narrow band of frequencies. The frequency at the centre of the band is controlled by the *FREQUENCY* control.

Now put various sounds into the Filter and sweep the *FREQUENCY* control from 0 to 10 to find their frequency components. Use as low a setting for the source device as possible because the Filter is most selective when its input signal is small. A pin at H3 enables you to analyse a sine-wave or ramp, and moving it to H4 enables you to analyse a square wave or ramp, or combination of the two. Although the audible results are not quite as clear as the graphs, the harmonic structure should be apparent. You can continue the experiment with external sources, using the Input Amplifiers and a pin at H8.

Much of the original research into the frequency components of sounds was carried out in the nineteenth century by Helmholtz. He did not have the advantages of modern electronic equipment, and he made acoustic filters which serve the same purpose as the SYNTHI Filter in the experiment above, using the actual sounds rather than their electrical equivalents. His filters, now called *Helmholtz resonators*, were designed for maximum selectivity, but almost any container with one or two holes in it functions as a filter as you can verify by speaking into a tin-can or along a tube. Unlike an acoustic resonator, which works at one frequency only, the SYNTHI Filter's frequency can be adjusted over the whole audible range.

An important advantage of the sound spectrum graph over the waveform graph is its ability to show the characteristics of noise. "Noise" is a colloquial term used for any sound which is not deliberately organised for a purpose, as speech and music are. But it also has a technical sense which we will define here: *noise* is a sound composed of all frequencies. We also make a distinction between *white noise* which contains all frequencies equally, and *coloured noise* which contains more sound at some frequencies than others. As with "spectrum", the words are chosen by analogy with light: a mixture of light of all frequencies (colours) is white. Like any sound, noise has a waveform, but it is not a constant waveform like those of the other sounds we have been considering, and it is very little use for studying the nature

of noise. But the spectrum of noise is useful, and in fact is the most important characteristic of noise.

Experiment 6

Connect the Noise Generator of the SYNTHI to Output Channel 1 with a pin at A7.

Adjust the Noise LEVEL control so you can hear the noise. With the COLOUR control central, the noise is white (make sure you have OUTPUT FILTER Channel 1 set to 5); turning the COLOUR control to the left emphasises low frequencies, and turning it to the right emphasises high frequencies. See Fig. II-8.

Note: the Noise Generator requires 10 - 30 seconds to warm up after the SYNTHI has been switched on.

Experiment 7

Adjust the Filter to give a narrow passband as in Experiment 4. Connect the Noise Generator through the Filter to Output Channel 1 (H7, A10).

A "band" of noise can now be selected by moving the FREQUENCY control of the Filter. The width of the band can be altered by moving the RESPONSE control, but remember that above a certain setting of RESPONSE, the Filter will resonate, generating an additional pure note in the middle of the band of noise. See Fig. II-9.

CHAPTER III: TECHNIQUES OF ELECTRONIC MUSIC

In this Chapter we will look at the origins and history of Electronic Music, and the technical developments that lead to the SYNTHI, and then show how these principles are incorporated into the SYNTHI.

Electronic Music is almost as old as electronics; within a few years of the invention of the radio valve, Dr Theremin was experimenting with an instrument played by a performer who moved his hands around two aerials, and M. Martenot was building early versions of a keyboard instrument called the "Ondes Martenot". Both instruments are built in modified form and used today. The invention which had the most profound effect on electronic music was the tape-recorder. Although gramophone records had been in existence for a long time, and some of the pioneers of electronic music had made considerable use of them, a more flexible medium was needed. With tape, it became possible to edit, mix, superimpose and perform other important processes very much more easily than with discs.

The problem now became one of *control*: a well-equipped studio of the nineteen-fifties (and there were very few) might have an assortment of oscillators and filters, but very limited techniques for "performing" on them. A composition would be realised very laboriously by recording the required sounds, often one note at a time, and editing the resulting tapes until the composition was complete.

The next step was to make more devices to control the existing ones. The flexibility of a modern studio is determined less by the sound-producing devices themselves than by the controlling equipment. The SYNTHI uses the powerful technique of *voltage control*, and also an entirely new and sophisticated method of *sequencing*, or playing a number of notes in succession without intervention from the player. The most advanced modern studios use a digital computer, which may be regarded in this application as a very advanced sequencer.

Voltage Control

So far, with one exception, the signals which we have used in the SYNTHI have been analogous to sounds; that is, the signal could be taken to an Output Amplifier at any stage in the process and be heard. The exception was at the end of Experiment 2 when we watched the Meter move in time with the Attack/Decay cycle: with the pin at B11 we can read the voltage, but if we move it to A11, connecting Trapezoid to Output Channel 1, there is no sound. This is not because the signal is of a different kind, but because its frequency is too low to be heard. Although such signals are no use for making sounds, they may be very useful for controlling other devices, and they are called *control voltages* for this reason.

Experiment 8

Connect Oscillator 1 to the Filter (H3), the Filter to the Envelope Shaper (D10), and the Envelope Shaper to Output Channel 1 (A12). Use the Meter to display the Trapezoid voltage (B11) and adjust the Envelope Shaper timing controls for a medium cycle about 2 seconds long. Turn the TRAPEZOID LEVEL control up so that the Meter sweeps across most of the scale (the Meter switch should be set to CONTROL VOLTAGE). Now use the Trapezoid voltage to affect three different variables of the sound:

1. Use the Trapezoid to control the frequency of the Oscillator with a pin at I11. The pitch swoops in time with the Attack/Decay cycle. The FREQUENCY control of the Oscillator still has an effect, but now it only alters the average frequency.
2. The Trapezoid will control the filtering if the pin is moved to N11. Once again, it is the frequency of filtering which is controlled and this can be very useful in instrumental simulation.
3. The Trapezoid can be used to control the Output level. Since the Envelope Shaper is already controlling the Output level of

Channel 1, put a pin at C10, which takes the Filter output to Channel 2, bypassing the Envelope Shaper. Put another pin at P11, and Channel 2 level will fluctuate because the Trapezoid is now controlling its level. Note that the two channels are "out of phase" - when one is loud, the other is soft, and vice versa. In fact, with suitable adjustment of the timing controls, it is possible to make the sound appear to move from one side to the other. It may be necessary to set the TRAPEZOID LEVEL control to 8 or 9 to achieve this effect. See Fig. III-1.

In some electronic music systems, great care is taken to distinguish signals and controls. In the SYNTHI there is no such distinction, and the result is a considerable economy in wiring and number of devices. Any voltage may be used either to produce a sound or to control a device. All we can say is that voltages with very low (often subsonic) frequencies will more often be used for control, and voltages of higher (audible) frequencies will normally be used as sounds.

If you look at the top of the Patchboard, you will see that the inputs are divided into two sections, *Signal Inputs* and *Control Inputs*. So far we have used the Signal Inputs mostly, and we will now look at the Control Inputs in more detail; we will refer to a Control Input by the letter in the corresponding at the bottom of the Patchboard, I, J, etc. The Control Inputs do not provide any new functions, but they do enable the manual controls to be altered automatically. They do not override the Manual Controls, but add their own effect to them.

I,J,K: Voltages in these columns control the frequencies of the three Oscillators. The range of voltage control is greater than that provided by the knob. Oscillators 1 and 2 have the same sensitivity, so they can be set to a given interval, say a fifth apart, and that interval will maintained as the frequencies are changed by the same voltage. There are no "forbidden" combinations on the SYNTHI - an Oscillator can control its own frequency, or the Oscillators can be connected in a ring, controlling one another.

- L: This column controls the decay time of the Envelope Shaper; +2V lengthens the longest decay time (knob set to 10) by about 50% to 25 seconds.
- M: This column controls the proportion of reverberated signal mixed with direct signal when the Reverberation Unit is used. It can be switched on and off rapidly, permitting unusual effects.
- N: This column controls the frequency of the Filter. As with the Oscillators, the effective frequency range of the Filter is increased by voltage control.
- O,P: These two columns control the signal level of Output Channels 1 and 2. When they are in use it will often be necessary to adjust the OUTPUT LEVEL controls to balance the channels correctly.

Experiment 9

In this experiment we give a simple example of the use of each voltage control input. It is a useful exercise to build more elaborate patches from them, trying to discover more interesting sounds. We do not give control settings in detail, but leave it to you to discover useful combinations of controls.

- (a) Patch: A12, D3, I4. Oscillator 2 is used to control the frequency of Oscillator 1. A good vibrato is obtained with Oscillator 2 controls at 3, 5, 0, 1.
- (b) Patch: A12, D4, J7, J11. Oscillator 2 is controlled by Noise and Trapezoid. The Noise LEVEL control can be adjusted to give an almost pure tone or a wide range of coloured noise.
- (c) Patch: A13, E5, F3, I5, I15, K16. This is a more complicated patch than we have used before, and it is illustrated on a Dopesheet. Oscillator 3 is controlled by the Joystick, and is ring-modulated with Oscillator 1 to produce "plucked" sounds.

Description:

Name: EXPERIMENT 9 (c)

Sheet:

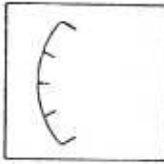
Patch:

Start:

End:

Notes

External Connections:



U	S

FILTER/OSC

RINGMOD

OSCILLATOR 1

ENVELOPE SHAPER

OSCILLATOR 2

External Patch:

CONTROLS

REVERBERATION

OSCILLATOR 3

[illegible]

2	3	de	re	ti
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INPUT LEVEL

NOISE GENERATOR

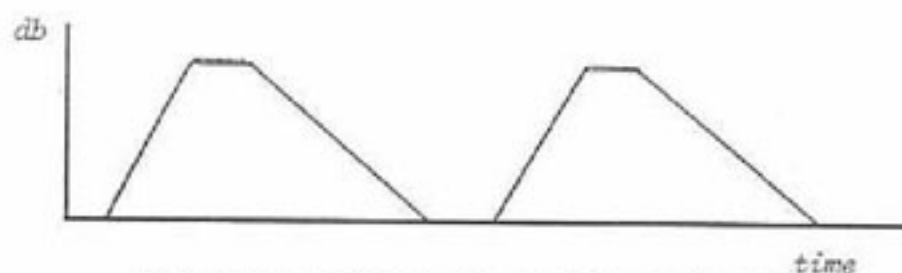
OUTPUT FILTER

CHANNEL 1 - - - OUTPUT - - - CHANNEL 2

A B C D E F G H I J K L M N O P

The other Joystick output controls the frequency of Oscillator 1 (I15) so both the repetition rate and the pitch of the notes can be altered in "performance". The Joystick gives good manual control of any two parameters of the sound, and a use can be found for it in practically every patch.

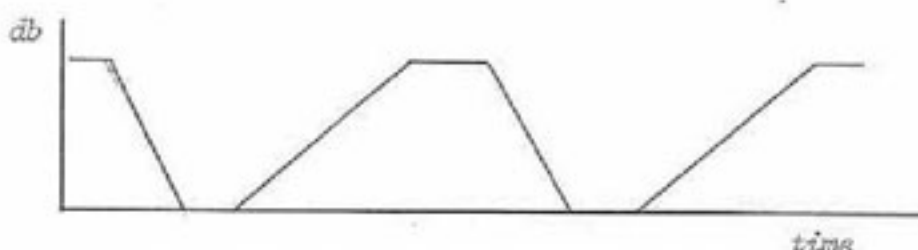
- (d) Patch: A12, D7, L6. Set the Envelope Shaper controls to 0, 0, 5, 0, 0, 10 and the Noise COLOUR to 5 and LEVEL to 10. Oscillator 3 (initially at 5, 6, 0, 6) controls the Decay time and hence the cycle duration of the Envelope Shaper.
- (e) Patch: A14, E5, F3, G13, M5.
Oscillator 1: 5, 5, 8, 0
Oscillator 3: 5, 5, 5, 0
Ring-modulator: 10
Reverberation: 7, 10
Correctly set up, this patch can produce a sound resembling a short roll on a pitched drum. Without the pin at M5, which is controlling the reverberation mix, it is merely a blur. This patch illustrates the value of controlled reverberation.
- (f) The patch is shown on the Dopesheet. It illustrates another way of making a dull sound from an unmodified Oscillator into an interesting musical timbre, in this case by controlling the Filter frequency with Oscillator 3 and the Trapezoid.
- (g) Patch: A3, C12, D3, O11. This patch moves the sound (in this example it is Oscillator 1, but it could be any sound) from one channel to the other. It depends on the fact that the Trapezoid output is inverted, so that the Channel controlled by the Trapezoid is loud when the output of the Envelope Shaper is loud, and vice versa.



(a) Envelope Shaper Signal Output Level

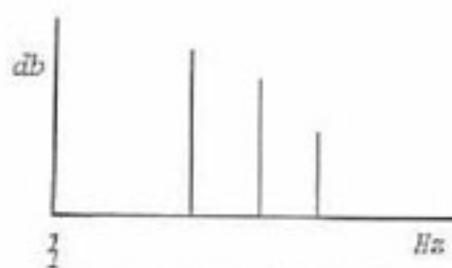


(b) Envelope Shaper Trapezoid voltage

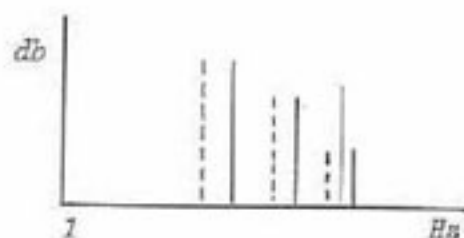


(c) Signal level of Amplifier controlled by Trapezoid

Fig. III-1: Panning with the Envelope Shaper and Trapezoid voltage



(a) Spectrum of C (261Hz)



(b) Spectrum of E (330Hz)



(c) Spectrum of Ring Modulation products of C and E

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Fig. III-2: Spectra of C, E and their Ring Modulation products

Modulation

When a signal is modified in some way by another signal, we say that the first is *modulated* by the second. The most common forms of modulation in conventional music are small fluctuations in the amplitude or frequency of a note. Both kinds of fluctuation generally have a frequency of 5 - 10Hz and a proportion of 2 - 5%. Amplitude modulation is called *tremolo* and frequency modulation *vibrato* (these terms are sometimes confused). Experiment 9(a) showed how vibrato is obtained with the SYNTHI, using Oscillator 2 to modulate Oscillator 1. This can be made into a demonstration of tremolo by moving the pin at I4 to O4, so that Oscillator 2 is controlling the level of Output Channel 1.

Ring-modulation is a third kind of modulation which is often used in electronic music. The signals fed to a ring-modulator are multiplied, and the result is much richer in overtones than either of the input signals. To see why this is, consider two pure tones *A* and *B*, with amplitudes *a* and *b*, frequencies *f* and *g*, so

$$A = a \cdot \sin(2\pi ft)$$

$$B = b \cdot \sin(2\pi gt)$$

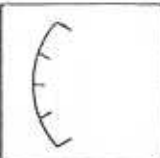
If these are ring-modulated, the result is a signal *C*, where

$$C = A \cdot B$$

$$= ab \cdot \sin(2\pi ft) \cdot \sin(2\pi gt)$$

$$= \frac{1}{2}ab \cdot \cos\{2\pi(f-g)t\} + \frac{1}{2}ab \cdot \cos\{2\pi(f+g)t\}$$

That is, the result is a sound with two frequency components of equal amplitudes and frequencies which are the sum and difference of the original frequencies. If we put in more complex sounds with more than one frequency component, all the frequency components of each sound will interact in this way and many new frequencies will be produced. Fig. III-2(a) shows the spectra of two sounds each with two overtones, the fundamentals being at middle C and at E a third above. Fig. III-2(c) shows the spectrum of the result of ring-modulating these two sounds, and it has eighteen different frequency components.

SYNTHI Dopesheet E.M.S.(London) Ltd.		Description:		External Connections	
Name: EXPERIMENT 10				RING MOD 7	
Sheet:					
Patch:		FILTER/OSC 5		9	
Start:		7			
End:					
Notes:					
OSCILLATOR 1		8 5 6 0		ENVELOPE SHAPER	
OSCILLATOR 2		7 5 5 5		0 2 8 0 9	
OSCILLATOR 3		7 5 1 4			
NOISE GENERATOR		OUTPUT FILTER		REVERBERATION	
6		5		5 9	
CHANNEL 1 - - OUTPUT - - CHANNEL 2		6		INPUT LEVEL	
				7	
				RANGE	

External Patch	SIGNALS						CONTROLS						16			
	out	meter	2out	env	A m.	B	reverb	filter	1 osc	2 freq	3	decay		reverb	filter	1 out lev
out chan	1															
osc 1	2															
osc 2																
osc 3																
noise																
inputs 1																
inputs 2																
filter																
trapezoid																
env signal																
ring mod																
reverb																
stick																

A B C D E F G H I J K L M N O P

The result of ring-modulation is sometimes harsh because many of the new frequencies are not musically related to each other. However, instruments such as bells and chimes have resonances which also are not related in a musical way, and the ring-modulator can be used to simulate them; although it is usually still necessary to filter out some of the new sounds which it introduces.

Experiment 10

The patch and control settings for this experiment are shown in the Dopesheet. The Keyboard is used to provide a pitch voltage at Input Channel 1, and Input Amplifier 1 LEVEL should be adjusted accordingly. The tuning of Oscillators 1 and 2 is critical since very small changes in their relative pitches will produce sounds of widely varying timbre. The sound can be made to give deep bell sounds or lighter chime sounds by adjusting the Filter FREQUENCY.

Reverberation

When a source in a large room or hall stops producing sound, the sound we hear does not cease abruptly, but continues to echo round the room for a short time. This is called *reverberation*. In assessing the reverberation time of a particular room, we make a loud abrupt sound, and measure the time taken for it to decay to one-millionth of its original intensity. The time depends on frequency, so the measuring device must use a burst of white noise (sometimes a pistol shot) or a sound with many frequency components.

A classroom has a reverberation time of at most 1 second: a longer time would make speech hard to understand. A small hall for chamber music has a reverberation time of 0.9 to 1.7 seconds, and a large hall for orchestral music may be from 1.5 to 2.5 seconds. Cathedrals and large churches have very long reverberation times - as much as 8 seconds - and are suitable only for choral and organ music written for them.

Electronic music is in some ways like instrumental music recorded in a room with no reverberation at all. It may be interesting musically but it is dull and lifeless to listen to. Having constructed electronic analogues of acoustic oscillators and filters and so on, to complete the picture we ought to simulate reverberation. But there is a difficulty because reverberation involves long delays, which are difficult to produce with purely electronic techniques. There are various ways of getting round this problem. The best is to use a soundproof room with a loudspeaker and a microphone inside it; the loudspeaker makes noises, and the microphone records the noises directly, and also the reverberation from the walls of the room. A good alternative is to use a large steel plate with *transducers* on it; one transducer is similar to a loudspeaker but it makes the plate move rather than the air around it, and the other corresponds to the microphone but it is designed to pick up vibrations of the plate. The principle is that echoes from the edges of the plate are similar in nature to echoes in a room.

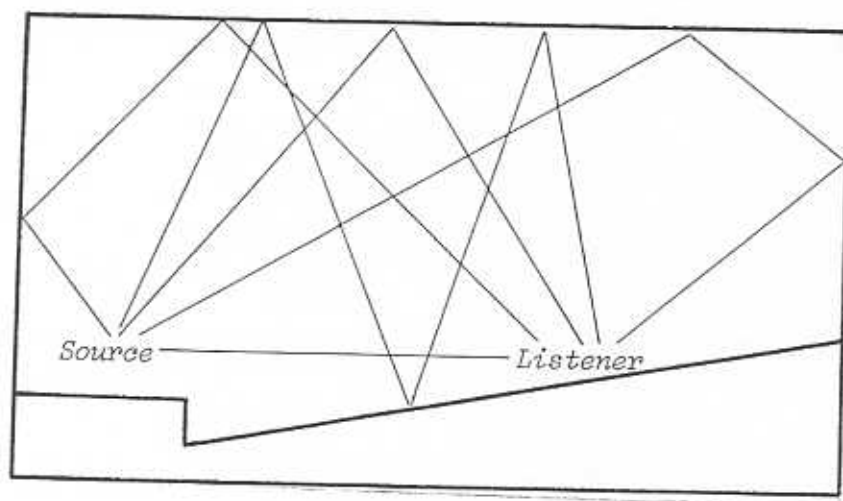


Fig. III-3: Reverberation in a room



Fig. III-4: Spring Reverberation Unit

Unfortunately, we could not build either a room or a large steel plate into your SYNTHI, but we did manage to fit in two small springs, and these provide an effective imitation of reverberation in a small space. Once again, two transducers are used and signals are sent along the springs by the first and collected at the other end by the second. Two springs are used rather than one to simulate the complex "multiple echo" nature of live reverberation; the delays are slightly different so that the echoes do not reinforce one another. In the SYNTHI, the springs have delays of approximately 25 milliseconds and 30 milliseconds, the time taken for sound to travel 23 feet and 27 feet respectively.

In the SYNTHI, the amount of reverberation can be altered by the Reverberation MIX control. A large amount of reverberation (MIX at 7 - 9) is a very dramatic effect but is seldom required. It is often much more effective to give a dry sound a little "life" with reverberation MIX at 3 - 5. It is usually best to patch the Reverberation Unit last in a chain of devices, since placing it early in the chain can make the final sound muddy, but sometimes this effect can be exploited, as is shown in Experiment 11.

Note: Unlike any of the other SYNTHI devices, the Reverberation Unit uses a mechanical link which is subject to external vibration. This has two effects: the first is that when reverberation is used, the SYNTHI is sensitive to mechanical shocks or vibration, which will be picked up by the springs and amplified. The other is that the sound of the Monitor loudspeakers may be picked up in the same way, resulting (at worst) in feedback and howling. It is therefore best to avoid high levels of reverberation when the internal loudspeakers are being used.

Experiment 11

The patch and control settings are shown in the Dopesheet. The sound is rather uninteresting with no reverberation (MIX at 0) but can be given a haunting quality by increasing the MIX until the optimum level is found.

This patch is likely to cause howling if the experiment is performed using the internal loudspeakers.

[illegible]

Level Controls and Mixing

Every device on the SYNTHI has an output level control. At first, it may seem unnecessary to have, say, four different level controls in a chain of four devices, but in fact they are all important. In order to achieve the best possible sound from a SYNTHI, it is important that at each point in a chain of devices the level should be correctly adjusted. It is not possible to define a set of hard and fast rules for setting levels, but we will give some general guides.

Firstly, the level should remain as constant as possible through the chain of devices. If at any stage the level is high, the next device may distort. Levels can be checked by patching each device in turn to either an Output Channel or the Meter.

Some devices require special treatment. The Filter will produce more dramatic effects if the signal fed to it has a very low level, and the Ring Modulator will "break through" (infact signals will go straight through to the output, usually, distorted) if the signal is larger than about 1.5V p-p. The Reverberation Unit is capable of giving very good results, but it should not be overloaded, especially at high settings of the MIX control.

A very low level at any stage should be avoided, since the next stage will have to be set at high gain which may add to the noise level.

Mixing is the combination of two signals into one. In Oscillator 1 and 2 the two waveforms are mixed, so each oscillator has two level controls but only one output. Other devices can be mixed in the SYNTHI by the simple expedient of putting several pins in one column. Control voltages as well as signals can be mixed, and this facility is often used, for example, when adding vibrato. Clearly, when two signals are mixed both the absolute and relative levels are important.

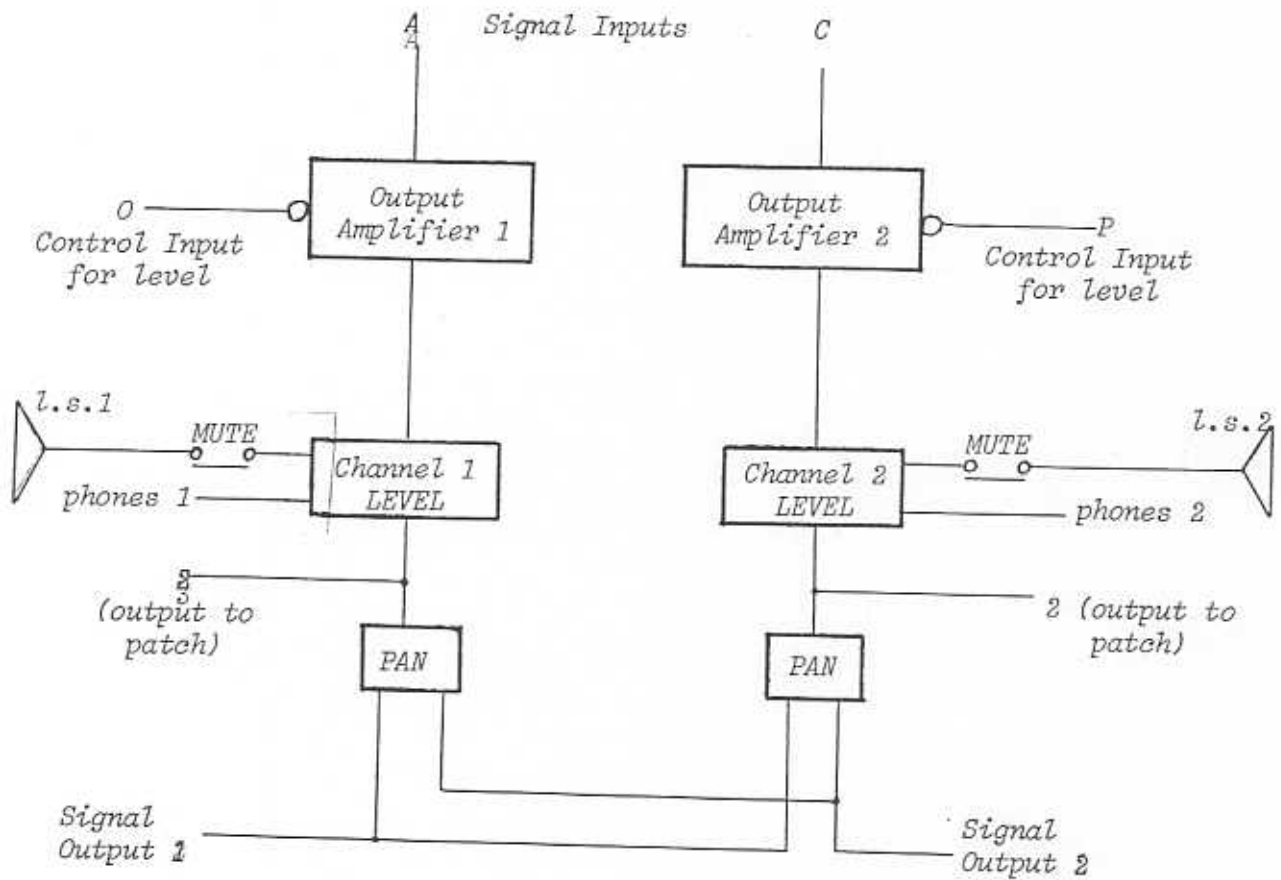


Fig. III-5: Block diagram of Output Amplifiers

Letters (A, C, O, P) and numbers (1, 2) are patchboard connections.

Panning

It is obvious that with two loudspeakers a sound can be made to come from one or the other. It is not quite so obvious that the sound can be made to come from any point in between them, by sending a proportion of the sound to each channel. In the early days of stereo recording, which depends on this fact, there was a great deal of dispute about whether it was necessary to have phase differences between the loudspeakers as well. The answer is complicated, but roughly speaking, our ears make good use of any information they can get, and dividing the sound proportionally works sufficiently well for most purposes. *Panning* (a word probably borrowed from the film-industry) means moving the sound from one side to the other. In the SYNTHI this can be done manually, using the PAN controls on the Output Channels, or automatically using the Envelope Shaper and Trapezoid (see Experiment 9(d)). The PAN controls, which enables two sounds to be moved independently between the loudspeakers, make the SYNTHI a true stereophonic machine, not simply a two channel machine.

A signal which is faded slightly and simultaneously given more reverberation seems to recede into the distance. Using this fact and simultaneously panning the signal gives effective three dimensional control.

Note: *The PAN controls are only operative when external amplifiers are used. To clarify this point, refer to Fig. III-5, which is a functional diagram of the Output Channels.*



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Output Filters

The Output Filters are incorporated in the Output Amplifiers, and they are a kind of "tone control". They should be used for making final adjustments to the sound, and also for matching the SYNTHI to power amplifiers connected to it or to the acoustics of the room.

CHAPTER IV: THE KEYBOARD AND SEQUENCER

Note: This chapter describes the use of the electronic touch-keyboard and the digital sequencer which are part of the SYNTHI AKS. It does not apply to the EMS keyboards DKO and DK1, or to the EMS Sequencer 64.

As we have seen already, it is often easier to make a piece of electronic equipment than it is to devise a way of controlling it. However complex a synthesizer we make, performances on it are ultimately limited by the player's ability to turn knobs. We can only extend the power of the synthesizer by making it easier to play.

The SYNTHI AKS uses a touch-keyboard engraved with a piano-like scale of black and white notes. Although it is possible to play tunes on this keyboard as if it was a one-finger organ, we should think of it more as an additional device for providing two *control voltages* which may be connected to any of the voltage controlled devices in the SYNTHI. As a special case, we may choose to control the pitch of an oscillator and the loudness of an amplifier in order to play tunes, but this is only one application.

The *Sequencer* is a special form of storage unit. When the Sequencer is *recording*, any key touched will cause a control voltage and a time to be stored. Later, when the *sequence* is *played back*, each voltage reappears at the appropriate time after the beginning of the sequence. The Sequencer is rather like a tape-recorder (which also has record and playback functions) but as it is completely electronic, the pitch and speed of a sequence can be changed independently, and there is no "rewind". The Sequencer can only be used to record notes from the touch keyboard; it will not record other instruments or voltages. The largest number of events that can be stored is 256.

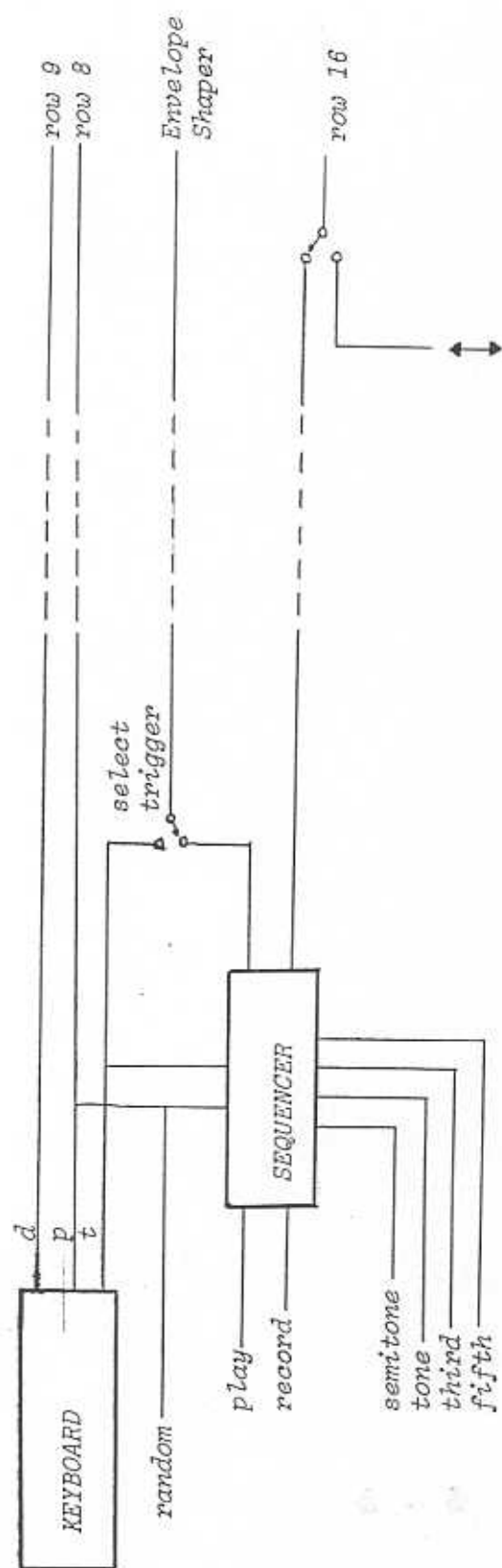


Fig. IV-1: Simplified Block Diagram of the SYNTHI Keyboard and Sequencer

Functional Description

Fig.IV-1 is a simplified block diagram of the Keyboard and Sequencer. The dotted lines represent the lead which joins the Keyboard unit to the SYNTHI proper.

The Keyboard generates three voltages: p is a voltage which depends on which key is touched, and is called the *pitch voltage*; d depends on how hard it was struck and is called the *dynamic voltage*, and t is a *trigger voltage* to fire the Envelope Shaper. The pitch about may be adjusted by the REALTIME PITCH SPREAD control, and determines the interval (tone, semitone, quartertone, etc) between each contact. The dynamic voltage is an inverted trapezoid which is normally used to control an output amplifier. The trigger is switched so that the Envelope Shaper can be controlled either by the Sequencer or by the keys being touch; this switch is on the extreme right of the Keyboard controls. The RANDOM key plays a note at random when it is touched and is connected to the pitch voltage line. The random note is selected from the notes stored in the memory. For example, suppose you record a sequence with C played 6 times, D played 3 times and E played once. Then the RANDOM contact will play C with a probability of 0.6, D with a probability of 0.3, etc.

When the Sequencer is recording, it receives pitch and trigger signals and passes them straight through so you can hear what you are recording. The SEQUENCER PITCH SPREAD control enables you to tune the Sequencer independently of the Keyboard. The Sequencer sends the direct or recorded pitch voltage to row 16 of the Patchboard so that sequences can accompany playing. (The vertical RANGE control must be switched to zero and only horizontal movements of the Joystick are effective when the Sequencer is used.) The contacts marked FIFTH, THIRD, TONE and SEMITONE raise the pitch voltage from the Sequencer when they are touched. The pitch change depends on the tuning, and it is only as stated when the tuning is chromatic (12 notes per octave); for other tuning, the FIFTH raises the pitch 7 notes, and the others 4, 2 and 1 notes respectively.

The Sequencer has a clock, and the duration of the sequence depends on the speed at which the clock is running. In order to achieve precise timing, you should run the clock at the fastest rate you can without the sequence being too short for your requirement. The duration is adjusted by the SEQUENCE LENGTH control on the left of the Keyboard, and the duration of the sequence is read from the Meter as described below. The SEQUENCE LENGTH control determines the *maximum possible length* of the sequence: you determine the *actual* length of the sequence by pressing the PLAY contact when you have finished recording.

The Sequencer provides another output which is connected to the Meter when the latter is switched to CONTROL VOLTAGE. This is a reading of the Sequencer clock; it is zero at the beginning of the sequence, and moves to the right, stopping at maximum deflection at the end of the sequence. The Sequencer is always recycling, whether it is in record or play mode, and the Meter indicates its position in the cycle. The Meter should be used to adjust the sequence duration before recording as described in the next experiment. The Meter can still be used to measure control voltages with reasonable accuracy since a pin in the Meter column will override the Sequencer output.

Experiment 12

This experiment is in two parts: (a) describes a simple application of the Keyboard alone, and (b) shows how the Sequencer is used. For simplicity, both experiments involve playing tunes.

- (a) To use the Keyboard alone, patch D3, A12 (Oscillator 1, Envelope Shaper, Output). We are using the pitch voltage from the Keyboard to control Oscillator 1 frequency, so put a pin at I8. Since the pitch voltage comes through Input Amplifier 1, turn INPUT LEVEL 1 to 7 and adjust the Oscillator to give a reasonable pitch and loudness. The Envelope Shaper switch on the Keyboard should be set to KEYBOARD. When you touch a contact, the ATTACK lamp will light, and if the Envelope Shaper is correctly set, you will hear a note.

To tune the Keyboard, touch contacts an octave apart alternately (not together - the Keyboard can only play one note at a time) and adjust the REALTIME PITCH SPREAD control until the two notes are an octave apart. Alter the Oscilloscope FREQUENCY and Envelope Shaper controls to obtain notes of different register and duration.

The dynamic voltage from the Keyboard is connected to row 9 (Input 2). An extra pin can be put at 09 for loudness control or at L9 for decay control. If you patch reverberation (remove A12 and substitute G12, A14) then you can control it from the Keyboard (M9). Play random notes as well: they can be interspersed in a melody if desired.

- (b) To use the Sequencer, patch D3, A12 again, and put the pitch voltage pin at I16. The ENVELOPE SHAPER control on the Keyboard should be turned to SEQUENCER (since this is where the trigger pulses will come from) and the vertical RANGE control should be set to zero (there is a switch at that position). Set the Meter mode switch to CONTROL VOLTAGE.

The Meter will be moving slowly across the scale to the right, and resetting at the end of its travel. This represents the maximum length of your sequence, and you should adjust the SEQUENCE LENGTH control on the Keyboard so that the time in which the Meter moves across is a little longer than the tune which you wish to play. Record your tune as follows:

Touch RECORD. The Meter will move to zero, indicating that the sequence has started.

Play your tune.

Touch PLAY. This will make the Sequencer play the tune from the beginning and then repeat it indefinitely. If you use the Sequencer to play a repeated pattern (such as an Alberti bass figure) it is important to touch PLAY quickly after the last note, or there will be a gap before the sequence starts again each time round.

Once stored, a sequence will play indefinitely. You can alter its pitch with the transposition contacts (FIFTH, THIRD, etc) or Oscillator 1 FREQUENCY, and its tempo with the SEQUENCE LENGTH control.

Some practice is required to use the Keyboard and Sequencer effectively. The Sequencer stores the beginning and end of each note, so the length of time for which you hold a note down should be used expressively. It is important to set the SEQUENCER LENGTH control correctly: incomplete storage or apparent timing errors may occur if it is too long or too short. A good length for recording a sequence is 25 seconds. After recording, a 25 second sequence can be varied from 2 seconds up to as long as 4 minutes. You should leave a small gap after the last note in the sequence, or you may record a "blip". When playing, the PLAY contact can be used to reset the sequence to the beginning, and a sequence can be cued by releasing the PLAY contact. Fortunately, it is very easy to re-record a sequence by touching the RECORD contact again, so start with a simple tune and try to obtain a perfect recording of it.

Tuning the Keyboard

When you touch a note of the Keyboard, you are actually generating a number. The number is in 5-bit binary form (in which form it is very easily stored by the Sequencer) and it is led to a *Digital to Analogue Converter* which turns it into a voltage. This voltage has one of 32 values, depending on which note you touched, and the range of voltage is altered by one of the SPREAD controls. The effect of this is that rather than tuning every note, or one particular note, we tune by *interval*. When the Keyboard is tuned conventionally, the voltage range is about 0.85V for the whole Keyboard, or 0.32V/octave which is correct for SYNTHI Oscillators 1 and 2. If we tuned it to 1.6V, we would have a 5 octave range and each note would be a whole tone apart; if we tuned it to 0.42V we would have a quarter tone scale covering a tenth. Once we have established a "spread" or range, for the Keyboard, we only have to tune one particular note to a particular frequency for the whole keyboard to be in tune.

The procedure for tuning the SYNTHI Keyboard is as follows: Tune the Sequencer by patching A3, I16 and adjusting the SEQUENCER PITCH SPREAD control for a perfect octave (or other interval if you can hear it better). Then choose one note, for example concert A = 440Hz, play it, and adjust Oscillator 1 to the desired pitch.

The procedure for the Keyboard is essentially the same but for one point. The Keyboard pitch voltage is taken through Input Amplifier 1 to row 8 of the Patchboard and so the Keyboard range is effected by the level of the Input Amplifier. It is simplest to preset the INPUT LEVEL to 7 and then not to alter it so long as the Keyboard is in use. The Keyboard can now be tuned with the KEYBOARD PITCH SPREAD control in the same way as the Sequencer.

The next experiment shows you how to accompany a sequence. The Sequencer is used to control Oscillator 1, and the Envelope Shaper is driven by the Sequencer trigger. When a sequence has been recorded, the Keyboard can be used to accompany it using Oscillator 2. The dynamic voltage from the Keyboard is similar to the control voltage produced by the Envelope Shaper but without time controls; here it is used to control the level of Output Channel 2.

Experiment 13

The Sequencer uses Output Channel 1. The signal path is connected with pins at D3, A12 and the pitch control uses one pin at I16. On the SYNTHI, adjust Oscillator 1, the Envelope Shaper, and Output Channel 1, and set the vertical RANGE control to zero. On the Keyboard, set the ENVELOPE SHAPER switch to SEQUENCER. Record a sequence as in Experiment 12 and then turn Channel 1 LEVEL control down while you set up the accompaniment patch.

Connect Oscillator 2 to Output Channel 2 (C4) and adjust the levels so that the sound is just inaudible at mid-range frequencies. Add a pin at J8 for pitch control and another at P9 for dynamic control. Turn INPUT LEVEL 2 up to 10 (the Input Amplifier is amplifying the dynamic control signal) and notes touched on the Keyboard should be audible. The only control of the envelope of each note in this mode is the relative levels of Input Amplifier 2 and Output Amplifier 2, but it

is possible to obtain a range of sounds from a staccato "ping" to a legato "organ" sound. Now by turning up Output Channel 1 LEVEL again, you can accompany the sequence.

The timbre of either voice can be enriched by using other devices. You should set up the experiment as described and then improve the timbres with filtering, ring-modulation, reverberation, vibrato, etc.

Finally, we describe a few ways of using the Keyboard without using the Envelope Shaper or dynamic voltage.

Experiment 14

- (a) Patch A13, E3, F8, I8. This uses the ring-modulator to produce a short, staccato envelope. An interesting feature of this patch is that the loudness of each note depends on the interval between it and its predecessor.
- (b) Move E3 to E7. This is a "drum" sound which can be used as part of a patch for playing tunes.
- (c) The Keyboard and Sequencer can be used as a control for many SYNTHI patches. Try "playing" the Chimes described in Experiment 12, for example.

CHAPTER V: THE SYNTHI AND OTHER DEVICES

Amplifiers

Although the SYNTHI has internal amplifiers and loudspeakers, these are only intended for monitoring, and in demonstrations or concerts the SYNTHI should be connected to a stereo power amplification system. The SIGNAL OUTPUTS on the SYNTHI provide signals which are suitable for most amplifiers. They should be connected to the AUX or RADIO sockets on a pre-amplifier or integrated amplifier, or they may be connected directly to the line-input of a power amplifier.

Amplifiers and loudspeakers are designed to produce speech and music. You should always bear in mind that the SYNTHI can produce sounds which do not occur in speech or conventional music, particularly large signals at very high and very low frequencies, which can damage loudspeakers. If the loudspeaker is distorting badly or rattling, you should turn the levels down or disconnect the SYNTHI immediately.

Fig. V-1 shows a SYNTHI connected to a stereo amplification system.

Tape Recorders

Although the SYNTHI is an excellent instrument for live performance, a tape-recorder is essential if elaborate electronic compositions are to be realised, and two tape-recorders are better still. The outputs of the tape-recorder can be connected to the HI LEVEL inputs of the SYNTHI, and the SIGNAL OUTPUTS of the SYNTHI to the AUX or RADIO sockets of the tape-recorder. Fig. V-2 shows a SYNTHI connected to a stereo tape-recorder; later in this chapter we will look at ways of connecting more than one tape-recorder.

With a tape-recorder you can record several sounds in succession and then edit the tape to make a composition. By altering the speed of

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External Connections:

the tape-recorder between recording and playback you can change the speed and the pitch of sounds. If the tape-recorder has a separate playback head you can produce the interesting echo effects described in Experiment 15.

Experiment 15

Connect the AUX or RADIO input of the tape-recorder to the SYNTHI's SIGNAL OUTPUT, and the line-output of the tape-recorder (which must come from a separate playback head) to the HI LEVEL INPUT.

- (a) *Patch: A12, G8. Any sound presented on Input Channel 1 (for example, a note from the Keyboard) will be given a combination of reverberation and simple echo which is more effective than either separately.*
- (b) *This is a fairly complicated set up and we show it in a Dopesheet. Each time the echo returns from the tape, it is ring-modulated by a sine-tone from the Filter. Simple primary sounds are best, and the Dopesheet shows short notes provided by Oscillator 1. This patch is capable of rather beautiful sounds if some care is taken to get the levels adjusted optimally. The settings on the Dopesheet were obtained from a tape-recorder which had been lined up to give the same level at its output as it received at its input.*

Microphones and Instruments

As well as making its own sounds, the SYNTHI is able to make dramatic transformations of sounds fed to it. The MIC 600 ohm inputs are suitable for most air-microphones, contact-microphones and electronic instruments such as organs and guitars. Before connecting anything to the MIC sockets, turn the INPUT LEVEL controls to 0, then plug the device in and slowly turn up the INPUT LEVEL to a suitable setting. It is best to patch the Input Amplifier straight through to

the Output Channel (pins at A8, C9) and make sure that a clear undistorted sound is obtainable by direct amplification before attempting treatments. Alternatively, send the signal to the Meter (pin at B8, or B9, Meter switch to SIGNAL LEVEL) and adjust the INPUT LEVEL controls for a maximum indication of 4 - 7. If the signal is to be taken to the Filter or Ring-Modulator it should be set to a fairly low level (3 on the Meter) for best results.

For each new sound, it is best to try out each SYNTHI treatment by itself first, and then to try more elaborate transformations using several devices. Experiment 16 describes how to try out the effect of individual devices, and Experiment 17 demonstrates more complex effects.

Experiment 16

A mono signal is present at Input Channel 1 (row 8) and its level has been set as described in the text above.

- (a) *Patch: A10, H8. This filters the sound; the Filter may be controlled manually or automatically. For example, control it with Oscillator 3 (N5 or N6).*
- (b) *Patch: A12, D8. The Envelope Shaper can fade the sound in various ways. Try each of the following with various settings of the Timing Controls.*
 - Automatic: with OFF time at 7 or less the Envelope Shaper will recycle immediately.*
 - Manual: with the OFF time at 10, a new sound will be produced whenever the ATTACK button by the Joystick is pressed.*
 - Threshold Trigger: put a pin at C8, connecting the input signal to Output Channel 2, and put the right-hand loudspeaker switch to TRIGGER. The Envelope Shaper will recycle whenever the input signal reaches the triggering threshold. This can be used to give instruments an entirely different envelope without affecting their timbre.*

- (c) Patch: A14, G8. This adds reverberation to the sound. The optimum setting of the MIX control will depend on the nature of the input signal.
- (d) Patch: A13, E8. The Ting;modulator requires another input before it will do anything. To ring-modulate with a sine-wave, patch E10 and turn the Filter RESPONSE to 10 so the Filter resonates. Try the other Oscillators (E3, E4, E5 or E6), Noise (E7) and also ring-modulate the sound with itself (E8) and explain the result by referring to "Modulation" in Chapter III.

SYNTHI Peripherals

There are three special purpose units manufactured by E.M.S. for use with the SYNTHI or other electronic music equipment. These are described briefly below but come with their own detailed descriptions.

Pitch to Voltage Converter

This device is rather modestly named, since it will not only convert a pitch to a voltage, but will produce another voltage proportional to the amplitude of the signal (envelope following), and a trigger pulse (for the Envelope Shaper) as well. It is therefore possible to make the SYNTHI accompany a live instrument in tune (or in parallel harmony, an octave lower for example) and with correct rhythm, but with an entirely different timbre and envelope. Inverted voltage outputs are available so the SYNTHI can also accompany in contrary motion, or be loud when the live instrument is quiet.

Random Voltage Generator

The Random Voltage Generator can be used for a number of effects apart from playing random tunes. One of the problems of electronic music is that the sounds are often too regular, or mechanical, and a small randomly fluctuating voltage, used to provide vibrato or tremolo, for example, can do a lot towards making an exciting "live" sound.

Eight Octave Filter Bank

This unit comprises eight filters tuned to fixed frequencies an octave apart. It enables very subtle modifications to be made to the timbre of the sound, and is useful for eliminating unwanted components, such as hum or noise, without degrading the music content of the signal. The individual output from each filter is available which greatly increases the flexibility of the Filter Bank as well as making it more useful educationally.

A More Advanced Studio

You will have realised by now that the SYNTHI is a very versatile instrument. It is not an exaggeration to say that with a SYNTHI, a stereo amplifier, two tape-recorders, and some or all of the E.M.S. peripherals, you would have a comprehensive modern electronic music studio. There are many people involved in the production of electronic sounds for concerts, advertising, films and theatre sound effects with fewer resources than this.

It is virtually impossible to say exactly how all these devices should be connected, because this will depend on the particular application. All studios of any complexity therefore have a *patchfield*, which usually consists of a number of jack sockets, with plug-leads to connect them. E.M.S. recommend a *Patchboard*, as used in the SYNTHI, because they are more compact, more reliable, and easier to use. E.M.S. will provide a 16x16 matrix, which is sufficient for many applications, or a 60x60 matrix (as used on the SYNTHI 100) for larger studios. A 60x60 matrix is equivalent to a patchfield with 3600 sockets. Fig. V-3 is a block diagram of a very flexible studio build around a 16x16 Patchboard using the following equipment:

SYNTHI AKS

6 - 2 Stereo Mixer

Stereo Power Amplifier

2 Monitor Loudspeakers

2 Stereo Tape Recorders

E.M.S. Pitch to Voltage Converter

E.M.S. Random Voltage Generator

E.M.S. Eight Octave Filter Bank

2 Reverberation Units

The audio signal connections of each device are taken to the patchboard. The E.M.S. peripherals produce control voltages which can only usefully be taken to the SYNTHI. A four position switch selects the device to trigger the Envelope Shaper:

1. *Keyboard*
2. *Pitch to Voltage Converter*
3. *Random Voltage Generator*
4. *Attack Button*

Microphones can be connected to the mixer, the SYNTHI, or the tape-recorder. Two reverberation units are used in addition to the unit in the SYNTHI because reverberation is important in electronic music and external units, preferably using steel plates, will give superior results.

The photograph at the front of this book shows the large electronic music studio used by EMS for developing their products. Both the SYNTHI AKS and the larger SYNTHI 100 are shown, as well as the computers used for digital control.

CHAPTER VI: SYNTHI PRESTOPATCHES

In this Chapter we describe the three Prestopatches provided with the SYNTHI: how they work, and how they are used.

In each of the Patchboard pins there is a resistor. When you put a pin into the Patchboard, you are joining two devices in the SYNTHI with this resistor. All the devices connected to the Patchboard are also connected to the 32-way socket by the spare pin store, and so a plug can be pre-wired with resistors forming a patch and plugged in. A plug made in this way is called a *Prestopatch*.

A Prestopatch only contains the pin positions, not the settings of all the controls. Any Prestopatch is therefore incomplete without a Dopesheet which shows how the controls should be set. The Dopesheet should also show which controls should be changed to give particular effects.

E.M.S. will make up Prestopatches to your own specification. Send a Dopesheet with your patch clearly marked on it and we will provide a Prestopatch wired for your patch.

PRESTOPATCH 1: BATTLE

Naturally, the first thing to do when trying a Prestopatch for the first time is to plug it in, set the controls and listen to it. For the Prestopatch to be of any real value, however, you must understand how it works in order that you can alter the controls constructively rather than at random.

Experiment 17

Plug in the Battle Prestopatch and set up the controls as in the Dopesheet. Patterns of explosive sounds should emerge from

both channels; the texture can be altered with Oscillator 3 SHAPE control and the Joystick.

Try and understand from the Dopesheet how the Prestopatch works: the principle device chain is Noise, Filter (H7), Envelope Shaper (D10), Output 1 (A12), Reverberation (G12), Output 2 (C14). The Oscillators and Joystick are used to control Decay, Reverberation, Filtering, and Oscillator frequencies. Discover the function of each device by altering its controls.

You can always add pins to a Prestopatch, by inserting them in the Patchboard in the usual way. See if you can enhance the Battle Patch by introducing new controls or devices.

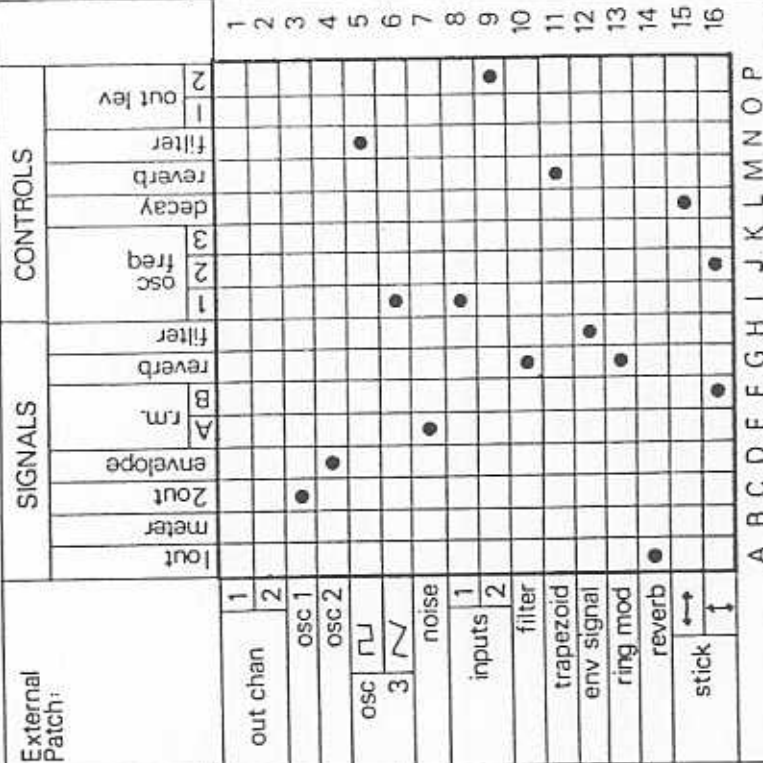
PRESTOPATCH 2: KEYBOARD

The Keyboard Prestopatch is used to obtain a variety of effects from the Keyboard and Sequencer; you will probably not use all the effects at the same time, and you will need to adjust the controls to get effects individually.

Experiment 18

Plug in the Keyboard Prestopatch and adjust the controls. The patch works in the following way: the Sequencer controls Oscillator 2 (J16) which is connected to the Envelope Shaper (D4), Filter (H12), Reverberation (G10) and Output Channel 1 (A14). Oscillator 3 controls the Filter to provide a tremolo effect (N5). The Sequencer also makes a drum sound using the Ring Modulator (E7, F16) which is mixed into the Reverberation Unit (G13). The Keyboard controls Oscillator 1 (C3) using the dynamic voltage to provide an envelope (P9). A vibrato is provided by Oscillator 3 (I6).

SYNTHI Dopesheet E.M.S.(London) Ltd.		Description: * Adjust these controls yourself		External Connections:	
Name: KEYBOARD PRESTOPATCH Sheet: Patch: Start: End: Notes:		 C S FILTER/OSC RINGMOD		 7 6 	
OSCILLATOR 1 4 5 10 10		ENVELOPE SHAPER 0 0 5 10 10		REVERB 6 7	
OSCILLATOR 2 Phase Mix 4 		OSCILLATOR 3 Filter Tremols 7 5 4 05		REVERBERATION 6 7	
NOISE GENERATOR 5 7 5 5		OUTPUT FILTER 5 7 5		INPUT LEVEL 7 7	
CHANNEL 1 -- OUTPUT -- CHANNEL 2 7 7		RANGE 7 0			



A B C D E F G H I J K L M N O P

First, tune the Sequencer by turning Channel 2 LEVEL down and adjusting SEQUENCER PITCH SPREAD. Then turn Channel 1 LEVEL down and use REALTIME PITCH SPREAD to tune the Keyboard. Finally, play both at once and tune them together with Oscillator 1 and Oscillator 2 FREQUENCY controls.

Now record a sequence as described in Chapter IV and then accompany it.

Find out which controls affect the sound in an interesting way; the following are the most important.

Oscillator 1 LEVELs - sound from Output Channel 2
Oscillator 2 LEVELs - sound from Output Channel 1
Oscillator 2 SHAPE - quasi-phasing effect
Oscillator 3 SQUARE - filtering
Oscillator 3 Triang - tremolo
Ring Modulator LEVEL - percussion
Trapezoid LEVEL - reverberation
Horizontal Joystick - decay

PRESTOPATCH 3: GUITAR

This Prestopatch is capable of providing a variety of effects from a single input. Although it is designed for use with an electric guitar, it can be used with any suitable input, including electronic organ or microphone.



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EXPERIMENT 19

It is best to establish levels first by connecting the guitar or other instrument to Input Channel 1 and patching it directly to the output (A8). For most instruments, the microphone input will be most suitable, but if the instrument overloads the microphone input (so the Input Amplifier LEVEL control has to be kept right down to avoid distortion) then the LINE input should be used.

Put in the Prestopatch and set up the controls. The effects possible with this patch include filtering, panning, reverberation, re-enveloping and octave-splitting, and the patch works as follows: the signal coming into the SYNTHI goes to the Ring Modulator (F8), and Filter (H8); the Filter goes directly to Output Channel (A10) and also to the Envelope Shaper (D10) and Reverberation Unit (G10). The Envelope Shaper and Reverberation Unit provide the outputs to the other channel (C12, C14), and the Ring Modulator is fed back into itself (A13, E13). Oscillator 1 LEVEL controls the panning together with Input Level 2 (the Prestopatch incorporates a connection from Oscillator 1 to Input Amplifier 2 which enables an inverted control voltage to be obtained). Oscillator 3 Square LEVEL controls reverberation, and Triang LEVEL controls filtering. The Trapezoid LEVEL controls also filtering, and the Envelope Shaper signal LEVEL controls the re-enveloped sound. The Joystick controls rate of panning horizontally and range of filtering vertically. The Ring Modulator provides an octave splitting effect, producing a note an octave lower than the one fed to it, and should be set up approximately as follows: Set Ring Modulator LEVEL at 8 and put the Bass control on the guitar to full.

This patch is capable of many dramatic effects. Experiment with it and note down settings which you find useful in performance. Here are the most useful controls:

<i>Oscillator 1 level</i>	<i>- width of panning</i>
<i>Oscillator 2 level</i>	<i>- variation in re-enveloping</i>
<i>Oscillator 3</i>	<i>- reverberation and filtering</i>
<i>Trapezoid level</i>	<i>- filtering synched with re-enveloping</i>
<i>Ring Modulator level</i>	<i>- octave splitting</i>
<i>Joystick horizontal</i>	<i>- rate of panning</i>
<i>Joystick vertical</i>	<i>- range of filtering</i>

CHAPTER VII - CARE AND MAINTENANCE

The Synthi could hardly be easier to maintain, because the solid state circuitry is designed to run well within its capacity, even under conditions of electrical misuse, and there are no mechanical parts except the Joystick. But the following general points may be helpful:

Jack Sockets.

These are of standard pattern, and extra jack plugs can easily be obtained. However there are some non-standard sizes on the market, and no plugs should be used if they are a very tight fit or on the contrary move too freely in the socket. They should push home with a firm click and have very little lateral play when in position.

Joystick.

The grease in the control slots may eventually dry out, particularly if the studio is kept in a warm place. To service, remove the bottom panel, and take off the red cover plate of the joystick. Carefully clean the slots and ball joint, and re-lubricate with a little Vaseline or silicone grease. Do not allow any grease to touch the potentiometers and take care not to cross-thread the self-tapping screws when replacing the cover.

Knobs.

If knobs become loose, slacken off the set screw reset at either maximum or minimum position, and tighten firmly. The spindles are nylon, and it is normal for the screw to bite in slightly for a firm fixing. If knobs are lost, we can supply more of the correct pattern, or in an emergency any knob suitable for a $\frac{1}{4}$ " spindle can be used. If knobs are wrenched so that the whole potentiometer becomes loose, attend to it at once before an internal wire is broken. Take off the back, hold the potentiometer from behind and tighten the nut firmly. If the wires are already displaced check that none are broken and resolder if necessary.

Meter.

If the pointer does not read zero when the studio is off, the zero can be adjusted with the small perspex screw at the bottom of the dial.

Panels.

Avoid any abrasive cleaner, and never use strong solvents like acetone, trichlorethylene or petrol (gasoline, benzine essence). The best cleaner is methylated spirits (alcohol), but paraffin (kerosene) or turpentine can be used, though they tend to leave an oily deposit and often an unpleasant smell. Use wax pencil to mark the panels if you wish to do so, rather than lead, particularly hard lead, pencil. Do not use ball point, which may leave a permanent indentation in the finish, or fibre-tipped pens, which often contain an indelible stain.

Storage.

Solid state devices dislike sustained heat. Never leave the VCS3 in a sealed car in summer sun, or in a similar situation. For long term storage choose a cool, dry position. If it is not used for a very long time (a matter of years) there may be trouble with electrolytic capacitors when it is switched on again, and the best insurance against this is to run the studio for a few hours at least every few months.

- DON'T connect unknown inputs with the input level controls wide open. Take particular care when connecting valve (tube) driven equipment to the VCS3.
- DON'T overrun devices constantly for hours on end if the outputs are not connected, this can happen without the user knowing. The safest course is to remove pins when the studio is finished with for the time being.
- DON'T connect unknown mains supplies without checking.
- DON'T interconnect jacks at the back of the studio without thinking carefully first. Particularly take care if you connect the Keyboard Jones Socket, because of the danger of short-circuiting the supply rails.
- DON'T grossly overrun the internal speakers, which may be damaged and give unreliable results thereafter. For any audience application, connect to external power amplifiers and concert speakers.
- DON'T continue to use the studio if unexplained noises, heat or smells occur. Stop and investigate.

HINTS AND SUGGESTIONS

If you set up a patch and there is no sound, check the following:

1. The power supply lamp is lit.
2. The speaker muting switches are up.
3. Levels on all devices, and the Output levels are sufficiently high (typically 5-8 for Sources, 7-10 for Treatments, but there are mp "rules"). Check input Amplifier levels if using external devices.
4. Oscillators are not running subsonically. (Below Frequency setting 3 for Oscillators 1 or 2, or 6 for Oscillator 3.)
5. Noise Generator needs 20-30 seconds to start up.
6. Envelope Shaper Off Time may be set too high for automatic recycling (try pressing the Attack Button.
7. Check the patch by swapping pins: the pins are reliable but occasionally dirt in the holes of the matrix board can prevent contact or damage a pin.
8. The Output Amplifiers can be cut off altogether by a large control voltage in rows O and P.

If you have tried all these things with no success, ascertain which device (or devices) is not working, and remove the appropriate circuit board (the location of the devices is explained above).



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